

Reinforcement learning with forecast-loss rewards for sensor scheduling under power constraints

Abstract

Under power constraints, the forecast value of a channel changes over time. Prediction-Driven Proximal Policy Optimization (PD-PPO) is a complete online scheduler whose training configuration combines a forecast-loss policy reward, behavior cloning pretraining, continuing advantage-weighted behavior cloning during PPO, and auxiliary future-condition classification. A feasibility check enforces steady-state and startup-peak power limits, while a minimum dwell-time constraint determines the executed activation vector. The scheduler input comprises observation histories, channel states, the previous executed activation vector, and simulated warning signals. A frozen primary forecaster supplies the multi-step forecast loss used to construct the training loss and evaluate policies offline; it is absent from online action selection.

After pilot/model-selection seeds 117–118, the primary analysis used event-balanced evaluation windows for 22 post-selection evaluation seeds (119–140). Across these seeds, the complete PD-PPO training configuration achieved lower macro-averaged normalized forecast loss than the validation-selected fixed-schedule baseline in every seed. The same configuration also achieved lower mean forecast loss than that baseline in every seed; the respective reductions were 8.0% and 10.6%. Both endpoints remained lower in every seed over the continuous 5,242-step scoreable test partition. The PD-PPO scheduler also had lower losses than the

AoI-priority, round-robin, random-feasible, and privileged one-step look-ahead reference using next-step test targets in every seed, and lower macro-averaged loss than the Double DQN training configuration in every seed. Confidence intervals for matched PPO training configurations with age of information (AoI) and uncertainty objectives, and for the warning-rule baseline, included zero. All six candidate actions satisfy both power limits in this $q = 1$ instance, so the optional-channel cardinality and minimum dwell-time constraints provide the active scheduling restrictions. Evidence is limited to a controlled simulation with informative warnings and does not isolate the reward, PPO update, warning signals, or training-only supervision.

Keywords: Sensor scheduling, Power constraints, Time series forecasting, Reinforcement learning, Adaptive sensing, Antarctic monitoring

Abbreviations

AoI, age of information; AWS, Antarctic automatic weather station; DRL, deep reinforcement learning; PD-PPO, Prediction-Driven Proximal Policy Optimization; PPO, proximal policy optimization; SOC, state of charge.

1. Introduction

Power constraints can prevent a multi-sensor system from operating all measurement channels simultaneously. The scheduler must therefore select channel subsets over time while satisfying the steady-state and startup-peak power limits and the minimum dwell-time constraint. Round-robin scheduling follows a predefined cycle. AoI- and uncertainty-based policies use measurement freshness or estimated uncertainty to guide channel selection. These decision criteria do not directly

quantify how the observation sequence produced by a schedule affects multi-step forecast loss.

The forecasting value of an optional channel varies with the operating condition. Particle measurements are most informative during particle events. Flux events favor snow mass flux measurements, and thermal events favor measurements of snow-surface temperature. A newly activated channel may not affect the forecast immediately. Some instruments require warm-up, and the forecaster uses a window of recent observations. The minimum dwell-time constraint also limits how soon the active subset can change. A decision based only on the current step may therefore fail to minimize forecast loss over the following time steps.

This study presents Prediction-Driven Proximal Policy Optimization (PD-PPO). The PD-PPO scheduler denotes the complete scheduling method; the PD-PPO policy is its trainable categorical neural policy. The policy reward contains a training-loss term derived from multi-step forecast loss. Each candidate action index identifies an activation vector, and a feasible-action mask removes indices whose vectors violate either power limit. The policy selects a proposed action index and its associated vector. After an action change, the execution rule retains the previous executed vector for the minimum dwell time of $D_{\min}$ steps.

Figure 1 summarizes the experimental timeline, the online constrained-scheduling loop, and the offline computations used for policy training.

At each time step, the scheduler input includes recent sample-and-hold values and observation masks together with channel-level runtime information. It also includes the previously executed channel subset, budget and time features, and simulated warning signals. The current condition label weights the training loss, while a future-condition supervision label constructs the supervised target action

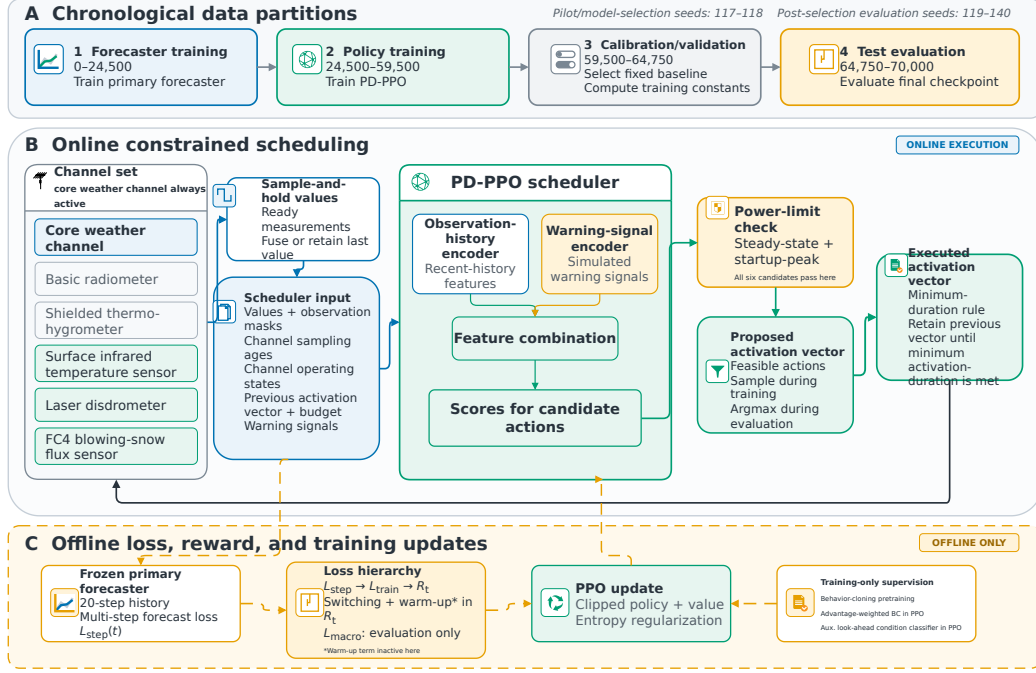


Figure 1. Chronological data partitions and the PD-PPO scheduling loop. Panel A presents the forecaster-training, policy-training, calibration/validation, and test partitions in chronological order; training constants and the supervised-action mapping are determined from calibration/validation data before policy optimization. Panel B shows how the scheduler input is converted into a proposed channel subset and then into the executed subset under the feasibility and minimum dwell-time constraints. Panel C presents the frozen-primary-forecaster computation of forecast loss, its conversion to training loss and policy reward, and the PPO and supervised updates.

and trains the auxiliary future-condition classifier. Neither label is a deployment-time scheduler input. During evaluation, current condition labels group the reported results. The frozen primary forecaster remains fixed throughout policy training and evaluation. The policy selects a proposed channel subset, and the minimum dwell-time constraint determines whether it can be executed or the previous subset must remain active.

The evaluation uses a controlled simulation of cold-region monitoring mo-

tivated by Antarctic blowing-snow sensing. The six-channel system keeps the meteorological core channel active and permits at most one of five optional channels to operate. Candidate actions are checked against the power and startup constraints. In this $q = 1$ instantiation, every candidate action satisfies both limits, so the optional-channel cardinality and minimum dwell-time constraints provide the active restrictions. Particle, flux, and thermal events favor different optional channels. The primary comparison uses a fixed schedule selected on the calibration/validation partition; a warning-rule baseline using the same deployment-available simulated warnings is the practical adaptive comparator.

The primary analysis covers 22 post-selection evaluation seeds. Across these 22 seeds, the complete PD-PPO training configuration has lower macro-averaged normalized forecast loss than the validation-selected fixed-schedule baseline in every seed. The same configuration also has lower mean forecast loss than that baseline in every seed, with reductions of 8.0% and 10.6%, respectively. Both endpoints remain lower in every seed over the continuous 5,242-step scoreable test partition. The PD-PPO scheduler also has lower losses than the AoI-priority, round-robin, random-feasible, and privileged one-step look-ahead reference using next-step test targets in every seed. Its macro-averaged loss is lower than that of the Double DQN training configuration in all 22 seeds and lower than that of the ridge-forecaster-selected fixed schedule in 21. A separate 24-seed descriptive aggregate retains the two pilot/model-selection seeds and the previously reported 10.1% and 7.9% reductions. The confidence intervals for comparisons with the two alternative scalar-objective PPO training configurations and the warning-rule and privileged true-label baselines include zero. The true-label baseline uses current and future condition labels that are unavailable during deployment, so it does not

define an upper bound for feasible sequential policies.

The study makes three contributions.

1. The study formulates sensor scheduling under power constraints as a sequential decision problem. It gives sufficient conditions under which condition-dependent channel selection improves on fixed policies that always select one optional channel and shows that forecast loss, AoI, and objectives based on covariance trace can rank feasible schedules differently.
2. The PD-PPO scheduler uses a complete training configuration that combines a forecast-loss policy reward, behavior cloning pretraining, continuing advantage-weighted behavior cloning during PPO, and auxiliary future-condition classification. A feasibility check enforces steady-state and startup power limits, and the minimum dwell-time constraint determines the executed subset.
3. The evaluation compares matched PPO training configurations with forecast-loss, AoI, and bounded diagonal uncertainty-proxy objectives and includes a Double DQN training-configuration comparison. Action-sequence analysis examines policy behavior; full-partition and alternative-forecaster analyses assess sensitivity to temporal support and forecaster choice.

2. Background and Related Work

2.1. Resource-constrained sensing

Resource-constrained sensor scheduling has been studied extensively in estimation, communication, and energy-aware monitoring. Estimation-oriented formulations minimize state-estimation error or posterior covariance ([Shi et al., 2011](#); [Ahdab et al., 2025](#)). Energy-harvesting censoring policies optimize the utility

of transmitted measurements (Fernández-Bes et al., 2015), while freshness-driven policies use age of information (AoI) or age of incorrect information (Kaul et al., 2012; Jonah et al., 2026). Reinforcement learning has also been applied to tracking and belief-state scheduling (Qu et al., 2022; Alali et al., 2024) and to delay-aware remote estimation with broader scheduling objectives (Tran et al., 2026). These objectives directly match evaluation criteria based on measurement freshness or current uncertainty. The present forecasting problem retains the resource constraints and uses forecast loss as the scheduling objective. A channel receives greater scheduling value when its observations reduce forecast loss over the evaluation horizon.

A forecast-loss objective is particularly relevant for heterogeneous sensing systems. A meteorological core channel can provide inexpensive and broadly useful measurements, while optional channels may provide their greatest forecasting benefit near specific physical events. The validation-selected fixed-schedule baseline therefore provides a meaningful comparator. The evaluation compares observation-dependent scheduling with fixed and dynamic baselines and with a privileged true-label baseline.

2.2. Adaptive sensing and learning-based scheduling

Forecast-oriented sensor scheduling is related to active perception and partially observable decision making. In these settings, an agent selects observations sequentially to improve a downstream task (Bajcsy et al., 2018; Lauri et al., 2023). Related theoretical work studies adaptive observation selection under partial observability (Golovin and Krause, 2011). These formulations support policies that respond to available observations. They do not address how forecast-oriented channel selections should be executed under steady-state power limits, startup-peak

Table 1. Position of this work relative to nearby scheduling and sensing literatures.

Literature line	Typical optimization target	Scheduling structure	What this paper adds
State-estimation and AoI scheduling	Covariance, estimation error, information age	Sensor selection under communication or energy limits	Multi-step forecast loss is the objective.
Active perception and adaptive observation selection	Task information under partial observability	Observation actions selected from partial observations	A mandatory meteorological core channel, limited optional-channel capacity, and baselines that use information unavailable during deployment.
DRL-based adaptive sensing and sensor steering	Information gain, coverage, data acquisition utility, or monitoring reward	Learned adaptive policies, often with soft or task-specific constraints	A finite candidate action set enforces the mandatory-core and optional-channel-capacity rules, and a feasible-action mask checks the steady-state and startup-peak power limits.
Time series forecasting models	Forecast accuracy after the observation stream is given	Forecasting model for a given observation stream	A frozen primary forecaster evaluates every schedule; its parameters are fixed during policy training and evaluation.

power limits, and a minimum dwell-time constraint.

Deep reinforcement learning has been used for resource-constrained adaptive sensing and sensor steering (Murad et al., 2020; Ogbodo et al., 2026), informative path planning (Wei and Zheng, 2020), and constrained optimization with delayed rewards (Pendyala et al., 2024). Its sequential objective can account for consequences that emerge over multiple time steps. Explicit feasibility enforcement is required when the sensing system cannot execute arbitrary actions. The PD-PPO policy is categorical over candidate action indices, and the scheduler applies a feasible-action mask at each time step. The mask removes indices whose candidate activation vectors violate the steady-state or startup-peak power limit.

2.3. Forecasting models for schedule evaluation

Forecasting models can quantify the downstream prediction loss associated with an observation sequence produced by a sensor schedule. Relevant architectures include the Temporal Fusion Transformer, temporal convolutional networks, and iTransformer (Lim et al., 2021; Bai et al., 2018; Liu et al., 2024). The primary forecaster is trained on the forecaster-training partition, and its parameters are held fixed throughout policy training and evaluation. During policy training, it provides the multi-step forecast-loss term used to construct the policy reward. During test evaluation, it measures the forecast loss of the observation sequence produced by each schedule. The alternative-forecaster analysis applies a ridge forecaster to the same observation sequences and reselects the fixed-schedule baseline on the calibration/validation partition. This sensitivity analysis quantifies the same comparison under a second forecasting model.

2.4. Antarctic blowing-snow monitoring

Antarctic automatic weather station (AWS) networks and blowing-snow monitoring provide the application context for the simulation. Antarctic AWS networks supply long-term atmospheric and surface observations, while their spatial instrument coverage remains uneven (Wang et al., 2023). Field measurements and numerical models have been used to study drifting-snow occurrence and transport, atmosphere–snow interactions, and snow particle properties (Amory, 2020; Sharma et al., 2023; Monrad-Krohn et al., 2026). The controlled simulation represents this setting with one mandatory meteorological core channel and five optional channels. At most one optional channel can operate at a time. For each seed, the PD-PPO scheduler and all baselines are evaluated on the same simulated ground-truth time series.

3. Problem Formulation

3.1. Sensor scheduling formulation

Let $x_t \in \mathbb{R}^d$ denote the simulated physical state at time step t . The state contains meteorological and blowing-snow variables, including wind speed and direction, air temperature, relative humidity, air pressure, solar radiation, snow surface temperature, snow particle statistics, and snow mass flux. Let $a_t \in \{0, 1\}^m$ denote the activation vector applied to the m logical channels at time step t . Each logical sensing channel is a simulation-level unit with associated observations, power demand, startup demand, and warm-up behavior.

The sensing system contains a mandatory meteorological core channel and a set of optional channels. The core channel remains active because it supplies the minimum measurements needed by the downstream forecast task. At most one

Table 2. Notation used for scheduler inputs, actions, and constraints.

Symbol	Meaning
x_t	Simulated physical state at time step t
o_t	Scheduler observation built from observation histories, channel modes, channel sampling ages, the previous executed activation vector, budget/time features, and simulated warning signals
$\mathcal{A}$	Candidate action set satisfying the core-channel and optional-channel capacity rules
K	Number of candidate activation vectors in $\mathcal{A}$
n_{opt}	Number of optional channels, $ \mathcal{S} $
$\mathcal{F}_t$	Feasible activation-vector set satisfying the steady-state and startup-peak power limits at time step t
r_t	Remaining required activation time for the previous executed activation vector
$a^{(k)}$	Candidate activation vector indexed by k
$\mathcal{I}_t$	Feasible action indices $\{k : a^{(k)} \in \mathcal{F}_t\}$
u_t	Proposed action index sampled or selected from $\mathcal{I}_t$
$\bar{a}_t = a^{(u_t)}$	Proposed activation vector
a_t	Executed activation vector after the minimum dwell-time constraint

Table 3. Notation used for forecast losses, supervision, and policy training.

Symbol	Meaning
L_{step}	Per-step multi-step forecast loss with component-wise clipping at 10
L_{train}	Condition-weighted, calibration/validation-normalized training loss used in the policy reward
L_c	Event-type-specific forecast loss for event type c
s_c	Median calibration/validation event-type-specific forecast loss across candidate fixed channel subsets for $c \in \{\text{particle, flux, thermal}\}$
s_{calm}	Median overall calibration/validation forecast loss across the same candidate fixed channel subsets; default normalization factor for calm time steps
L_{macro}	Macro-averaged normalized forecast loss with equal weights for the three event types
L_ρ	Weighted normalized forecast loss used in Proposition 1
$\mathcal{V}_{\text{AoI}}$	Observed variables whose ages enter the theoretical AoI objective; not necessarily the forecast-target set $\mathcal{Y}$
π_{label}	Theoretical privileged policy with access to condition labels
π_s	Fixed-channel policy that selects optional channel s
c_t	Current condition label used for training-loss weighting on the policy-training partition and offline loss computation and grouping on the test partition; never a policy or critic input at decision time
c_t^{sup}	Future-condition supervision label: the first non-calm condition label in $[t, t + 16]$, or calm if none occurs; used only for behavior cloning and auxiliary classification during policy training
ℓ_t	Indicator of a valid supervised target action; $\ell_t = 1$ at every PPO step in the reported stride-one configuration
η_t	Indicator of an available future-condition supervision label for the auxiliary classifier; $\eta_t = \ell_t = 1$ at every PPO step in the reported stride-one configuration
S_t	Normalized Hamming distance $m^{-1} \ a_t - a_{t-1}\ _1$; fraction of executed channel activation bits that change
W_t	Number of warm-up interruptions at time step t

optional channel can be active at each time step. The candidate action set is

$$\mathcal{A} = \left\{ a \in \{0, 1\}^m \mid a_{\text{met}} = 1, \sum_{i \in \mathcal{S}} a_i \leq 1 \right\}. \quad (1)$$

Given the previously executed activation vector a_{t-1} , the feasible set contains the candidate actions that satisfy the steady-state and startup-peak power limits.

$$\mathcal{F}_t = \left\{ a \in \mathcal{A} \mid \sum_i p_i a_i \leq B, \right. \\ \left. \sum_{i: a_{t-1,i}=1} p_i a_i + \sum_{i: a_{t-1,i}=0} \max(p_i, p_i^{\text{peak}}) a_i \leq B^{\text{peak}} \right\}. \quad (2)$$

Here, $\mathcal{S}$ denotes the optional-channel index set. The steady-state powers p_i and budget B use a common normalized unit. The startup-peak powers p_i^{peak} and budget B^{peak} use a second common normalized unit. An active channel contributes its steady-state power to the startup-power check. A newly activated channel contributes the larger of its steady-state and startup-peak power. Together with Equation (1), these constraints permit a fixed schedule to select one optional channel while preventing more than one optional channel from operating at the same time.

The finite candidate set is indexed as $\{a^{(1)}, \dots, a^{(K)}\} = \mathcal{A}$. At time step t , the feasible indices are

$$\mathcal{I}_t = \{k : a^{(k)} \in \mathcal{F}_t\}.$$

During training, the policy samples an index from $\mathcal{I}_t$. During evaluation, it selects the highest-scoring feasible index. Let u_t denote the resulting index and let $\bar{a}_t = a^{(u_t)}$ be the proposed activation vector. Let r_t denote the remaining required activation time for the previous executed activation vector. The executed activation vector is

$$a_t = \begin{cases} a_{t-1}, & r_t > 0, \\ \bar{a}_t, & r_t = 0. \end{cases} \quad (3)$$

The feasible-action mask restricts the policy distribution to $\mathcal{I}_t$ and enforces the steady-state and startup-peak power limits. The execution rule retains a_{t-1} until its minimum dwell time has been completed. Each executed activation vector therefore remains active for at least $D_{\min}$ time steps.

3.2. Conditions under which optional-channel adaptation reduces forecast loss

The sensing configuration with a mandatory meteorological core channel and at most one active optional channel is a special case of a broader capacity-constrained scheduling problem. Let $n_{\text{opt}} = |\mathcal{S}|$ denote the number of optional channels, of which the scheduler may select at most $q < n_{\text{opt}}$ at each time step. For an operating condition $c \in \mathcal{C}$, let $S_c^* \subseteq \mathcal{S}$ denote an optimal optional-channel subset, with $|S_c^*| \leq q$.

Let $\tilde{L}_c(\pi)$ denote the normalized forecast loss of policy π under condition c . The weighted normalized forecast loss is

$$L_\rho(\pi) = \sum_{c \in \mathcal{C}} \rho_c \tilde{L}_c(\pi), \quad \rho_c > 0, \quad \sum_{c \in \mathcal{C}} \rho_c = 1.$$

The empirical macro-averaged normalized forecast loss in Equation (7) uses $\tilde{L}_c(\pi) = L_c(\pi)/s_c$ and assigns equal weight to the particle, flux, and thermal event types.

Dynamic scheduling can reduce forecast loss when no fixed optional-channel subset is optimal for every condition with positive evaluation weight. The advantage requires positive excess forecast loss when a fixed subset is mismatched to the operating condition. Predictive information in the scheduler input can then support channel changes before or during a condition transition. The following proposition gives a sufficient condition for this advantage.

Proposition 1 (Dynamic value with one active optional channel). *Consider a scheduling problem that permits at most one active optional channel. All policies are subject to the same startup and minimum dwell-time constraints. For any $s \in \mathcal{S}$, let π_s denote the fixed-channel policy that selects s under every operating condition. Suppose that at least two positive-weight conditions have different optimal optional channels. The corresponding condition intervals are long enough for the theoretical privileged policy π_{label} , which has condition-label access, to complete feasible channel changes under the common execution constraints.*

Transition losses are included in the normalized condition-specific losses. For every fixed policy π_s and every condition c ,

$$\tilde{L}_c(\pi_s) \geq \tilde{L}_c(\pi_{\text{label}}).$$

For each positive-weight condition whose optimal channel differs from s ,

$$\tilde{L}_c(\pi_s) - \tilde{L}_c(\pi_{\text{label}}) \geq \Delta_c > 0.$$

Then every fixed policy π_s that always selects one optional channel $s \in \mathcal{S}$ satisfies

$$L_\rho(\pi_s) > L_\rho(\pi_{\text{label}}).$$

A proof is provided in [Appendix A.1](#), and [Figure 2](#) illustrates the sufficient condition. The proposition assumes that all policies follow the same startup and minimum dwell-time constraints. It does not provide a performance bound for the PD-PPO scheduler or any other implemented policy. The empirical privileged true-label baseline in [Supplementary Table S11](#) is not assumed to implement the theoretical policy π_{label} or to attain the minimum forecast loss.

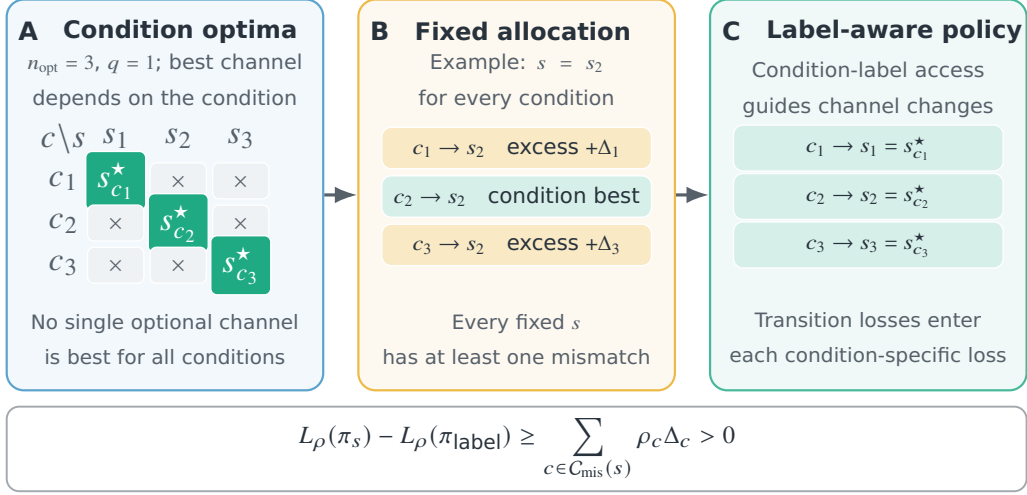


Figure 2. Sufficient condition for the value of adaptive scheduling with one active optional channel. Different operating conditions favor different optional channels, which causes every fixed policy that always selects one optional channel to incur excess loss for at least one condition. The theoretical privileged policy can change channels when the startup and minimum dwell-time constraints allow a feasible transition. The lower panel gives the resulting bound on weighted normalized forecast loss. The set $C_{\text{mis}}(s)$ contains the positive-weight conditions whose optimal channel differs from fixed channel s . The quantity Δ_c is the corresponding normalized excess loss.

Relation between the simulation and the proposition. The simulation was designed to instantiate condition-dependent channel optima of the type considered in Proposition 1. During particle, flux, and thermal intervals, the laser disdrometer channel, FC4 snow mass-flux channel, and infrared snow-surface-temperature channel, respectively, provide the most reliable measurements of the target variables emphasized for the corresponding event type. The event-type decomposition on the test partition in Table 7 evaluates whether the PD-PPO scheduler reduces loss under these event-specific measurement requirements and improves the macro-averaged normalized forecast loss. It does not verify the proposition’s inequality for every

π_s and the theoretical π_{label} in every condition. The proposition does not require every event type to have lower loss in every seed.

3.3. Forecast loss and macro-averaged evaluation across event types

The frozen primary forecaster receives a window of sample-and-hold values and observation masks and predicts selected target variables at horizons $k = 1, \dots, H$. Let $\hat{y}_{t+k,j}$ and $y_{t+k,j}$ denote the prediction and target, respectively, for variable $j \in \mathcal{Y}$ at horizon k . The condition label at time step t is

$$c_t \in \{\text{calm, particle, flux, thermal}\}. \quad (4)$$

Let w_k denote the horizon weight, α_{j,c_t} the condition-specific target weight, and σ_j the scale of target variable j . The per-step forecast loss is

$$L_{\text{step}}(t) = \frac{\sum_{k=1}^H \sum_{j \in \mathcal{Y}} w_k \alpha_{j,c_t} \min\{10, |\hat{y}_{t+k,j} - y_{t+k,j}| / \sigma_j\}}{\sum_{k=1}^H \sum_{j \in \mathcal{Y}} w_k \alpha_{j,c_t}}. \quad (5)$$

The target scales satisfy $\sigma_j > 0$, and all horizon and target weights are nonnegative. The denominator is positive for every scoreable time step. Each scale-normalized absolute error is capped at 10 before weighting and aggregation.

The experiment uses uniform horizon weights and the predeclared condition-specific target weights in Supplementary Table S4. The condition label determines which target weights are applied during policy training and evaluation. It is not included in the scheduler input. Losses for PD-PPO and all baselines are computed using the same target scales, target weights, and condition labels.

An overall average can be influenced disproportionately by event frequency and by differences in raw loss magnitude across event types. The macro-averaged normalized forecast loss addresses these effects by normalizing each event-type-specific loss with calibration/validation results from feasible fixed channel subsets

and assigning equal weight to the three event types. Let $c \in \{\text{particle, flux, thermal}\}$ index the event types, and let $\mathcal{T}_c$ denote the evaluated test time steps labeled as c . For a policy π , the event-type-specific forecast loss is

$$L_c(\pi) = \frac{1}{|\mathcal{T}_c|} \sum_{t \in \mathcal{T}_c} L_{\text{step}}(t; \pi). \quad (6)$$

For each event type c , the normalization factor s_c is the median event-type-specific forecast loss across the feasible fixed channel subsets evaluated on the calibration/validation partition. The macro-averaged normalized forecast loss is

$$L_{\text{macro}}(\pi) = \frac{1}{3} \sum_{c \in \{\text{particle, flux, thermal}\}} \frac{L_c(\pi)}{s_c}. \quad (7)$$

This metric is reported when $|\mathcal{T}_c| > 0$ and $s_c > 0$ for all three event types. Lower values indicate better forecasting performance. The metric compares policies across equally weighted event types after accounting for the median calibration/validation loss of feasible fixed channel subsets.

3.4. Non-equivalence of forecast loss, AoI, and covariance objectives

Forecast loss, AoI, and covariance trace evaluate different properties of a sensor schedule. Let π be a feasible scheduling policy and let $\gamma \in (0, 1]$ denote a discount factor. The same horizon T and temporal weights γ^t are used for all three objectives. The discounted forecast objective is

$$J_{\text{fcst}}(\pi) = -\mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t L_{\text{step}}(t; \pi) \right]. \quad (8)$$

For a separate model-based estimator with posterior covariance P_t , let $\mathcal{V}_{\text{AoI}}$ denote the observed variables whose ages enter an AoI objective. This set need not

equal the forecast-target set $\mathcal{Y}$. The AoI and covariance objectives are

$$J_{\text{AoI}}(\pi) = -\mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t \sum_{j \in \mathcal{V}_{\text{AoI}}} \text{age}_{t,j} \right], \quad J_{\text{cov}}(\pi) = -\mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t \text{tr}(P_t) \right]. \quad (9)$$

The three objectives can induce different orderings over the same feasible schedules. Such a difference can occur when sensor variables have unequal downstream forecast value, even if one schedule produces fresher observations and a lower estimator covariance.

Proposition 2 (Non-equivalence of forecast loss, AoI, and covariance objectives). *Forecast loss, covariance trace, and per-variable AoI objectives can induce different orderings over feasible sensor schedules. In particular, there exist a common horizon T , a common discount factor γ , and two feasible policies π_1 and π_2 such that*

$$J_{\text{AoI}}(\pi_1) > J_{\text{AoI}}(\pi_2), \quad J_{\text{cov}}(\pi_1) > J_{\text{cov}}(\pi_2), \quad (10)$$

and

$$J_{\text{fcst}}(\pi_2) > J_{\text{fcst}}(\pi_1). \quad (11)$$

Equivalently, π_1 has lower AoI and lower covariance trace, and π_2 has strictly lower forecast loss.

The proof in [Appendix A.2](#) takes the common horizon $T = 1$ and considers a single-target, single-horizon special case of Equation (5). A policy that refreshes the variable with greater influence on the forecast target has strictly lower forecast loss in the constructed example, while its total AoI and covariance trace are both higher. This ordering shows that forecast loss must be evaluated directly when sensor variables have unequal downstream forecast value. The AoI-priority, round-robin, and random baselines provide non-learned dynamic comparisons. The matched

scalar-objective PPO configurations replace the scalar scheduling objective while retaining the shared supervision and architecture.

The proposition establishes that the objectives can rank feasible schedules differently. It does not determine the performance ordering of policies learned from finite training data. The comparisons in Sections 4.5 and 6.3 examine whether the forecast-loss, AoI, and bounded diagonal uncertainty objectives produce measurable differences under matched PPO configurations. The empirical uncertainty proxy is not the posterior covariance trace used in the proposition.

4. PD-PPO Scheduler and Evaluation Protocol

4.1. Scheduler inputs, actions, and constraints

The PD-PPO policy is trained using the policy reward in Equation (14), whose forecast-loss term is derived from the multi-step forecast loss in Equation (5). The macro-averaged normalized forecast loss in Equation (7) is used for event-type-balanced evaluation and does not enter PPO training. At each time step, the scheduler receives recent sample-and-hold values and observation masks, together with the time since each channel’s last sampling opportunity. Its input also includes channel operating states, the previous executed activation vector, budget and time features, and simulated warning signals. The reported actor and critic do not receive the remaining activation time r_t explicitly; the environment enforces the lock after an action is proposed.

The finite candidate action set is listed in Table 4. A feasibility check removes actions that violate either the steady-state or startup-peak power limit and thereby forms the feasible-action mask. During training, the policy samples an index from the feasible action-index set. During evaluation, it selects the highest-scoring

feasible index. The resulting index specifies the proposed activation vector. The minimum dwell-time constraint in Equation (3) is then applied to determine the executed activation vector. Unless stated otherwise, all baselines use the same feasibility check and execution rule.

Table 4 gives the candidate channel subsets used by PD-PPO and the constrained baselines. Every subset includes the meteorological core channel, and each differs only in whether and which optional channel is active. The evaluated configuration therefore keeps one meteorological core channel active and allows at most one optional channel to operate.

Table 4. Candidate action set with a mandatory meteorological core channel and an optional-channel capacity of one.

Action index	Candidate channel subset	Power	Interpretation	Fixed-baseline selections
0	Meteorological core channel	0.25	Core channel only	0 of 22
1	Meteorological core + basic radiometer channels	0.33	Optional channel	0 of 22
2	Meteorological core + shielded temperature–humidity channels	0.43	Optional channel	0 of 22
3	Meteorological core + infrared snow-surface-temperature channels	0.74	Thermal specialist channel	9 of 22
4	Meteorological core + laser disdrometer channels	0.74	Particle specialist channel	2 of 22
5	Meteorological core + FC4 snow mass-flux channels	0.74	Flux specialist channel	11 of 22

Every candidate channel subset includes the meteorological core channel. The final column reports fixed baselines selected on the calibration/validation partition for the 22 post-selection evaluation seeds. The scheduling decision selects at most one optional channel; action 0 selects none.

In the reported $q = 1$ instance, the largest candidate steady-state demand is $0.74 < B = 0.75$, and the largest startup-peak demand is $0.92 < B_{\text{peak}} = 0.95$. Consequently, every one of the six candidate action indices belongs to $\mathcal{I}_t$ at every evaluated time step: neither power inequality excludes a candidate activation vector. The feasibility layer remains part of the method, but the active experimental restrictions are the optional-channel cardinality constraint and the minimum dwell-time constraint.

Let r_t be the remaining required activation time immediately before the action at time t . If $r_t > 0$, the environment executes $a_t = a_{t-1}$ regardless of the proposed $\bar{a}_t$. After execution, the update is

$$r_{t+1} = \begin{cases} D_{\min} - 1, & a_t \neq a_{t-1}, \\ \max(r_t - 1, 0), & a_t = a_{t-1}. \end{cases} \quad (12)$$

The first execution time step is therefore included in the $D_{\min}$ -step minimum. The reported values are $D_{\min} = 6$ and a maximum channel warm-up of two time steps. A newly activated channel cannot be deactivated before warm-up completes, so $W_t = 0$ structurally in this experiment.

4.2. Partial observations and scheduler input

The environment maintains sample-and-hold values without a model-based Kalman state estimate. Let $y_{t,i,j}$ denote a valid measurement of variable j provided by ready channel i at time step t , and let $R_{i,j}$ denote the configured measurement-noise variance. Define $\mathcal{O}_{t,j}$ as the set of ready channels that provide a valid measurement of variable j at time step t . For a scalar variable observed by at least

one ready channel, the fused value is

$$\tilde{x}_{t,j} = \frac{\sum_{i \in O_{t,j}} R_{i,j}^{-1} y_{t,i,j}}{\sum_{i \in O_{t,j}} R_{i,j}^{-1}}. \quad (13)$$

Wind-direction measurements are fused using a weighted circular mean. If $O_{t,j} = \emptyset$, the environment retains the previous value, $\tilde{x}_{t,j} = \tilde{x}_{t-1,j}$. An observation mask indicates whether each variable received a valid measurement at time step t . The scheduler and frozen primary forecaster receive the most recent 20 value vectors together with their observation masks. These masks distinguish new measurements from retained sample-and-hold values.

For scalar variables, Equation (13) assumes conditionally independent measurement errors with variances $R_{i,j}$. Under correlated channel errors, the rule provides a precision-weighted approximation. Computing the minimum-variance linear estimate would require the full measurement-error covariance matrix. The weighted circular mean for wind direction is defined separately from the scalar fusion rule.

At each time step, the environment first updates the channel operating states and identifies the ready channels. Measurement attempts are then performed for the ready channels. Valid measurements are fused into the current variable values, and variables without a valid measurement retain their previous values. The environment subsequently updates the observation masks and channel-level sampling ages before constructing the scheduler input.

For each channel, the scheduler receives its off, warming, or ready state, remaining warm-up time, and normalized sampling age. The sampling age measures the time since the channel's last sampling opportunity. It resets when an active channel becomes ready to sample, including cases in which the stochastic

measurement attempt produces no valid measurement. Variable-level AoI follows a different reset condition and counts the time steps since the variable’s most recent valid observation. The scheduler input further includes the previously executed channel subset, current power ratio, diurnal phase, and compact recent-history features.

The main policy does not receive an estimated covariance matrix. Covariance trace is used in Proposition 2 as an alternative scheduling objective. A diagonal predict–update uncertainty proxy is used only by the uncertainty-objective PPO configuration.

The scheduler receives ten additional features computed deterministically from the most recent 16 sample-and-hold value vectors and observation masks. The features summarize recent patterns associated with particle, flux, and thermal conditions. They also capture temporal trends and recent observation coverage. All ten features are derived solely from information available at decision time. Their complete definitions and clipping rules are provided in Supplementary Section S1.

Four simulated warning signals are also available to the scheduler at decision time. They represent particle, flux, thermal, and aggregate warning information and include simulated noise. Condition labels are excluded from the scheduler input. They are used to construct supervised information during training and to group results during evaluation. A learned gate conditioned on the aggregate warning signal combines the outputs of two feature encoders. The fused representation is then used to compute the candidate-action scores. The critic is implemented with a separate network. Supplementary Sections S1 and S5 provide the complete warning-signal construction and policy equations.

The implementation samples the policy at every training time step, including

Algorithm 1 PD-PPO policy training

Train the primary forecaster on the forecaster-training partition and freeze its parameters. Evaluate the candidate fixed channel subsets on the calibration/validation partition, compute s_{calm} and the event-type-specific normalization factors s_c , and define the condition-specific supervised target-action mapping.

Pretrain the policy for 10,000 behavior cloning steps using the supervised target actions derived from future-condition supervision labels.

for each PPO rollout of n_{steps} policy-training time steps **do**

 Initialize or continue a rollout on the policy-training partition.

for each time step t in the rollout **do**

 Construct o_t from the observation history, channel-level runtime features, the previous executed activation vector, budget/time features, and simulated warning signals.

 Apply the feasibility check to obtain $\mathcal{I}_t$.

 Sample $u_t \sim \pi_\theta(\cdot \mid o_t, \mathcal{I}_t)$ and set $\bar{a}_t = a^{(u_t)}$.

 Apply the minimum dwell-time constraint to obtain a_t .

 Advance the sensing environment with a_t and update the observation history.

 Evaluate the updated history with the frozen primary forecaster and compute R_t using Equation (14).

 Store the sampled proposed-action index u_t , its log probability, the reward produced after executing a_t , and the training-only supervision.

end for

 Update the policy and critic for n_{epochs} passes over the rollout using the combined objective in Equation (20).

end for

Save the final policy checkpoint after the prescribed PPO updates and freeze it for test evaluation.

when $r_t > 0$. If the lock overrides a proposal, the rollout buffer retains the proposed index u_t and $\log \pi_{\theta_{\text{old}}}(u_t \mid o_t, \mathcal{I}_t)$, whereas the environment transition, reward, and generalized advantage estimate are generated after executing $a_t = a_{t-1}$. The reported training protocol therefore uses proposed-action credit during locked time steps rather than an executed-action PPO likelihood. The logs do not retain the counterfactual proposals needed to estimate the override rate. This convention does not change the test losses of the frozen final policies, but those losses do not evaluate an executed-action PPO update.

4.3. Frozen primary forecaster and policy reward

The primary forecaster is trained before policy optimization. Its parameters are then frozen during policy training and evaluation. It receives the 20-step sample-and-hold value history and its observation masks. The time since each channel’s last sampling opportunity, runtime, warning signals, and the recent-history features above are provided only to the scheduler. During policy training, the frozen primary forecaster returns the multi-step forecast loss for the observation sequence produced by the current schedule. That forecast loss forms the training loss in Equation (16); the training loss and operational penalties form the policy reward in Equation (14). During test evaluation, the same frozen primary forecaster evaluates the observations produced by the PD-PPO scheduler and every baseline.

Supplementary Table S10 separates quantities available to the online scheduler from condition labels and future values used as training-only or evaluation-only information. This distinction separates the privileged true-label baseline from the PD-PPO scheduler.

The policy reward is the negative of the training loss plus the two operational

penalties used in the reported experiment.

$$R_t = - [L_{\text{train}}(t) + \lambda_{\text{sw}} S_t + \lambda_{\text{warm}} W_t], \quad (14)$$

where L_{train} is the condition-weighted training loss normalized using calibration/validation quantities, S_t is the normalized switching cost, and W_t is the number of warm-up interruptions introduced at time step t . With m logical channels, the executed-action switching cost is

$$S_t = \frac{1}{m} \|a_t - a_{t-1}\|_1. \quad (15)$$

This normalized Hamming distance is the fraction of executed activation bits that change between consecutive time steps. It is not a count of channel-switch events. Power and startup violations are excluded by $\mathcal{I}_t$, and the minimum dwell-time constraint determines a_t ; these constraints therefore do not enter the reward as residual violation terms. The reported evaluation separates mean forecast loss from the macro-averaged normalized forecast loss in Equations (5) and (7).

The training loss uses a condition-specific training weight and a normalization factor computed on the calibration/validation partition while retaining calm time steps and all three event types:

$$L_{\text{train}}(t) = \omega_{c_t} \frac{L_{\text{step}}(t)}{s_{c_t}}, \quad (16)$$

where $L_{\text{step}}(t)$ is the clipped, target-weighted, scale-normalized multi-step forecast loss returned by the frozen primary forecaster for the current observation history, and $\omega_c \geq 0$ is a condition-specific training weight. For each event type $c \in \{\text{particle, flux, thermal}\}$, $s_c > 0$ is the median event-type-specific forecast loss across candidate fixed channel subsets on the calibration/validation partition. The default factor $s_{\text{calm}} > 0$ is the median overall forecast loss across the same

candidates on that partition; it is used for calm time steps. Thus the denominator s_{c_t} in Equation (16) equals s_c when c_t is event type c and s_{calm} when c_t is calm. The particle, flux, and thermal weights are 1.55, 0.95, and 1.65, while calm time steps use unit weight. These weights and normalization factors are fixed before policy optimization and remain unchanged during policy training and evaluation. The current simulator-derived operating-condition label c_t is never available to the policy or critic at decision time. On the policy-training partition, it selects the condition-specific weight and normalization factor in Equation (16). On the test partition, it is used only for offline loss computation and grouped reporting. Test action selection requires the simulated warning signals but neither c_t nor a reward value.

Policy learning uses two additional supervised signals formed only during training. Define the future-condition supervision label c_t^{sup} as the first non-calm simulator-derived condition label in the window from t through $t + 16$, or calm when that window contains no event. This label selects a predefined feasible supervised target action. The meteorological core channel is paired with the shielded temperature–humidity channel in calm periods, the laser disdrometer channel for particle events, the FC4 snow mass-flux channel for flux events, and the infrared snow-surface-temperature channel for thermal events. These target actions supply behavior cloning labels, while c_t^{sup} supervises an auxiliary four-class future-condition classifier. Both signals are formed only on the policy-training partition. The policy and critic receive the simulated warning signals during policy-training rollouts and test evaluation. Neither c_t nor c_t^{sup} is available to the policy or critic at decision time. The target actions are first used for 10,000 steps of behavior cloning pretraining. During PPO, the advantage-weighted behavior

cloning term in Equation (21) and the auxiliary future-condition classification term in Equation (22) remain active.

The primary forecaster uses a TCN-style residual temporal convolutional architecture (Bai et al., 2018). The implementation has three levels, 64 channels per level, kernel size 3, and dropout rate 0.05. It is fitted on the forecaster-training partition before policy training. Input and target scales are fitted before policy updates, and the architecture is held fixed throughout policy training and evaluation. No forecaster weights are updated after the forecaster-training partition ends.

The forecaster is trained on observation sequences produced by full observation, fixed channel subsets, and dynamic baselines in the forecaster-training partition. This mixture exposes the model to dense histories and partial histories produced by different schedules before its parameters are fixed. It contains no calibration/validation or test observations. Rollout counts, reset conditions, and the complete training protocol are reported in Supplementary Section S2.

4.4. PPO objective

Let o_t denote the scheduler input, u_t the selected action index, $\bar{a}_t = a^{(u_t)}$ the proposed activation vector, a_t the executed activation vector, and $\hat{A}_t$ the advantage estimate. The policy defines a categorical distribution over the feasible action set,

$$\pi_\theta(u \mid o_t, \mathcal{I}_t) = \frac{\exp z_\theta(o_t, u)}{\sum_{v \in \mathcal{I}_t} \exp z_\theta(o_t, v)} \mathbf{1}\{u \in \mathcal{I}_t\}, \quad (17)$$

where $z_\theta(o_t, u)$ is the policy score for activation vector $a^{(u)}$ and $\mathcal{I}_t$ is the power- and startup-feasible index set. The activation vector containing only the mandatory meteorological core channel belongs to the candidate action set and satisfies both limits, so $\mathcal{I}_t$ is nonempty at every time step. PPO updates this distribution with the

clipped surrogate

$$\mathcal{L}_{\text{clip}}(\theta) = -\mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (18)$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(u_t \mid o_t, \mathcal{I}_t)}{\pi_{\theta_{\text{old}}}(u_t \mid o_t, \mathcal{I}_t)}. \quad (19)$$

The reported PPO loss contains the PPO surrogate, value loss, entropy regularization, and two auxiliary terms tied to the forecast training setup.

$$\mathcal{L}_{\text{PPO}} = \mathcal{L}_{\text{clip}} + c_v \mathcal{L}_V - c_H \mathcal{H}(\pi_\theta(\cdot \mid o_t, \mathcal{I}_t)) + \beta_{\text{BC}} \mathcal{L}_{\text{BC}} + \beta_{\text{cond}} \mathcal{L}_{\text{cond}}. \quad (20)$$

For labeled training examples, let $u_t^{\text{target}} \in \mathcal{I}_t$ denote the feasible supervised target action described above. The advantage-weighted behavior cloning term is

$$\mathcal{L}_{\text{BC}} = -\frac{\sum_t \ell_t [\hat{A}_t]_+ \log \pi_\theta(u_t^{\text{target}} \mid o_t, \mathcal{I}_t)}{\sum_t \ell_t [\hat{A}_t]_+ + \varepsilon}, \quad (21)$$

where $[x]_+ = \max(x, 0)$ and $\ell_t \in \{0, 1\}$ marks examples with a valid supervised target. Let $\eta_t \in \{0, 1\}$ mark examples for which the future-condition supervision label is available to the auxiliary classifier. An auxiliary future-condition classification loss is applied to the auxiliary future-condition classifier. Let h_t be the policy representation, $q_\theta(c \mid h_t)$ the auxiliary probability of operating condition $c \in \{\text{calm, particle, flux, thermal}\}$, and c_t^{sup} the future-condition supervision label used only during policy training. The reported stride-one configuration constructs both supervised targets from the same c_t^{sup} and sets $\eta_t = \ell_t = 1$ at every PPO step. The general implementation constructs the indicators separately. The auxiliary future-condition classification loss is

$$\mathcal{L}_{\text{cond}} = -\frac{\sum_t \eta_t \log q_\theta(c_t^{\text{sup}} \mid h_t)}{\sum_t \eta_t + \varepsilon}. \quad (22)$$

Supplementary Table S1 lists the full protocol and training hyperparameters.

4.5. *Reward-function and algorithm comparisons*

The AoI-objective and uncertainty-objective PPO configurations retain the same PPO policy and critic architecture, scheduler input, candidate action set, feasibility check, chronological partitions, condition-specific weights, supervised target actions available only during training, and auxiliary future-condition classification loss, but replace the scalar forecast-loss objective. The AoI-objective PPO variant averages normalized observed-variable age over the observation history; the uncertainty-objective configuration averages a bounded diagonal predict–update variance proxy. Their recursions and bounds are given in Supplementary Section S3. Each scalar objective supplies the corresponding rollout reward, from which the value targets and advantages are computed. Replacing it therefore changes the PPO surrogate, critic target, and the positive-advantage weights in $\mathcal{L}_{\text{BC}}$. These are matched PPO training configurations with different scalar objectives, not isolated scalar-reward ablations.

The Double DQN training configuration uses a dueling network and the same feasible-action mask, state representation, reward, action set, constraints, policy-training partition, and test windows as the PD-PPO scheduler. Online and target networks implement the Double DQN target over feasible next actions, with three-step returns and experience replay. This configuration does not use behavior cloning pretraining, the continuing advantage-weighted behavior cloning term, or the auxiliary future-condition classifier. Its complete settings are listed in Supplementary Table S1.

4.6. *Chronological data split and baselines*

Each simulated ground-truth time series is split into forecaster-training, policy-training, calibration/validation, and test partitions. The calibration/validation

partition computes s_c and s_{calm} , defines the condition-to-action mapping used to construct supervised targets on policy-training data, and selects the fixed-schedule baseline. Test windows are not used for forecaster training, policy updates, supervised-target generation, fixed-baseline selection, or normalization-factor fitting.

The calibration/validation interval follows the policy-training interval chronologically. Computationally, however, its normalization factors, condition-to-action mapping, and fixed-schedule selection are computed offline before policy optimization and then held fixed. Calibration/validation transitions are not used as policy-training rollouts.

The chronological split is summarized in Supplementary Fig. S1.

The fixed-schedule baseline tests whether adaptation is needed. The AoI-priority baseline, round-robin baseline, and random-feasible baseline provide non-learned dynamic comparisons under the same operating rules. The privileged one-step look-ahead reference using next-step test targets chooses the currently best feasible action; those targets are unavailable to the deployed scheduler. Its one-step objective omits minimum dwell-time consequences, warm-up, and observation age. It therefore does not isolate the value of sequential decision making. The warning-rule baseline maps simulated warning signals to channel subsets selected on the calibration/validation partition. The privileged true-label baseline receives the exact sequence $c_t, \dots, c_{t+16}$ at each decision; these condition labels are unavailable to the deployed scheduler. Neither privileged baseline is a guaranteed upper bound.

Supplementary Table S11 fixes the role of each baseline before the results section.

Exact decision rules, calibration/validation choices, tie breaks, and information

availability for these baselines are listed in Supplementary Section S6.

5. Simulation and Evaluation Setup

5.1. Simulated sensing system

The evaluation uses a controlled simulation of a cold-region sensing system motivated by Antarctic automatic weather stations (AWSs). The final experiment uses six logical channels, consisting of one mandatory meteorological core channel and five optional channels. Table 5 lists the channels and their normalized scheduling costs. A platform rendering is retained in Supplementary Fig. S2 as physical motivation, while the channel table defines the logical-channel configuration evaluated in the experiments.

The simulation keeps the meteorological core channel active, and at most one optional channel can be active. Core weather measurements remain available while the optional-channel capacity is allocated among logical channels whose forecast value changes across event types.

5.2. Simulated time-series generation

Each seed generates a simulated ground-truth time series with wind, temperature, humidity, pressure, radiation, snow-surface temperature, particle statistics, and snow mass flux. The simulation model uses reference distributions from the Antarctic AWS network and blowing-snow statistics. The meteorological variables are parameterized from the AntAWS compilation (Wang et al., 2023). Event and particle ranges draw on numerical modeling of atmosphere–snow interaction, Antarctic drifting-snow observations, and cold-region particle measurements (Amory, 2020; Sharma et al., 2023; Monrad-Krohn et al., 2026). Blowing-snow events are produced by a semi-Markov process coupled to wind exceedance.

Table 5. Logical channels and normalized scheduling costs.

Idx	Logical channel	Main observed variables	p_i	p_i^{peak}	Warm-up
1	Meteorological core channel	wind, air temperature, relative humidity, pressure	0.25	0.30	0
2	Basic radiometer channel	solar radiation	0.08	0.10	0
3	Shielded temperature–humidity channel	air temperature, relative humidity	0.18	0.22	0
4	Infrared snow-surface-temperature channel	snow surface temperature	0.49	0.62	1
5	Laser disdrometer channel	particle diameter, particle velocity, simulated particle latent variable	0.49	0.62	2
6	FC4 snow mass-flux channel	snow mass flux	0.49	0.62	1

The meteorological core channel is mandatory. At each time step, the scheduler may activate at most one optional channel and may leave all optional channels inactive. The infrared snow-surface-temperature channel, laser disdrometer channel, and FC4 snow mass-flux channel are event-specific channels for thermal, particle, and flux event types, respectively; the basic radiometer channel and shielded temperature–humidity channel are additional optional channels. Condition labels are simulator-derived metadata used only for training, calibration/validation, or offline evaluation; they are not direct channel measurements.

The simulation contains particle, flux, and thermal event types. Each event type changes a different part of the simulated physical state and is observed most reliably by a different event-specific channel. The laser disdrometer is used for particle events, the FC4 snow mass-flux channel for flux events, and the infrared snow-surface-temperature channel for thermal events. Event balance, targets weighted by event type, and the optional-channel capacity make all three event types matter in the forecast loss. One fixed optional channel can be effective on average, but no single optional channel covers the full forecast objective.

The scheduler input also contains simulated warning signals for the three event types. These noisy pre-onset signals are assumed available at decision time; their construction and distinction from condition labels are specified in Supplementary Section S5. They are simulator-generated and do not constitute a validated field warning system.

Eight prespecified validation checks assess the simulated series, covering meteorological structure and blowing-snow behavior (Aloni et al., 2025). They include wind autocorrelation, event frequency, agreement between simulated and observed distributions of scalar variables, event durations, the flux–wind relationship, particle-size–wind correlation, and the wind-speed power spectrum. All eight criteria pass their prespecified acceptance rules; the full table and simulation-validation figure are reported in [Appendix B](#). These simulation validation checks document the simulation construction and remain separate from test policy selection.

5.3. Metrics, seeds, and statistical summaries

The primary evaluation uses eight event-balanced evaluation windows of 512 time steps each. A prespecified rule balances event types and emphasizes transport periods. The rule uses event coverage and transport intensity from the simulated

ground-truth time series, not any policy loss, and is applied identically to the PD-PPO scheduler and every baseline. Mean forecast loss averages Equation (5) over the resulting 4096 time steps. The two co-primary endpoints are mean forecast loss and the macro-averaged normalized forecast loss in Equation (7). Normalization factors are computed from candidate fixed channel subsets on the calibration/validation partition, so the loss is specific to this simulation model. The primary forecaster architecture and training setup are described in Section 4.

Continuous full-partition evaluation examines performance beyond the event-balanced windows. The same final policy checkpoints and the fixed channel subsets previously selected on the calibration/validation partition start fresh continuous rollouts and are evaluated at every test step with a complete eight-step future target. This gives 5,242 scoreable test time steps per seed, covering [64750, 69992); only the last eight time steps are excluded because their future targets are incomplete. The policies, fixed channel subsets, and normalization factors fitted on the calibration/validation partition are unchanged, and no schedule is reselected from this evaluation.

The primary analysis reports 22 post-selection evaluation seeds. Pilot/model-selection seeds 117 and 118 support a prespecified two-variant architecture comparison using their selected test-window metrics. The selected configuration is then held fixed for the 22 post-selection evaluation seeds (119–140). For each seed, the same simulated ground-truth time series is used for the PD-PPO scheduler, the fixed-schedule baseline, AoI-priority, round-robin, and random schedules, and all reported baselines. The validation-selected fixed-schedule baseline is listed before test evaluation and then uses a constant feasible channel subset without fallback rotations.

The two pilot/model-selection seeds were used for a prespecified comparison between the reported warning-conditioned policy and a variant with an additional deterministic feature encoder. The former was selected before the remaining seeds were run. Across the 22 post-selection evaluation seeds (119–140), the complete PD-PPO training configuration achieved lower macro-averaged normalized forecast loss than the validation-selected fixed-schedule baseline in every seed (mean paired difference 0.0838; 95% percentile bootstrap confidence interval for the mean paired difference [0.0708, 0.0967]). The separate 24-seed descriptive aggregate retains the pilot seeds. The simulation parameters, target weights, baseline definitions, and reported training settings were held fixed for this expansion.

The main protocol hyperparameters are listed in Supplementary Table S1. Fixed channel subsets and calibration/validation losses for each seed are listed in Supplementary Table S9.

The alternative-forecaster analysis uses a ridge forecaster fitted on the same forecaster-training partition for each seed. The ridge forecaster selects its own fixed-schedule baseline using calibration/validation windows and then evaluates the unchanged PD-PPO and fixed-baseline observation sequences. Within this analysis, no policy, action sequence, or forecaster is updated from its test results. The current operating-condition label c_t is used offline to select the same predeclared target weights for every baseline and groups the completed losses for the macro-averaged normalized forecast loss.

Seed statistical summaries use paired differences because PD-PPO and each baseline are evaluated on the same simulated ground-truth time series. For a seed s , the paired difference is comparator loss minus PD-PPO loss; a positive paired difference favors the PD-PPO scheduler. Alongside mean paired differences and

counts of seeds with lower loss, the results report medians and 95% percentile bootstrap confidence intervals for the mean paired difference over seeds.

5.4. Action-sequence analysis

The analysis uses the concatenated executed-action sequences from the eight event-balanced evaluation windows. It reports executed-action entropy, consecutive-action-pair entropy, condition–executed-action mutual information, mean normalized Hamming distance between adjacent executed masks, and optional-channel activation frequency by operating condition. A sequence is classified as fixed-like when one executed activation mask accounts for at least 95% of actions and as simple-cycle-like when the top three masks account for at least 85% of actions and the best periodic match up to lag 64 reaches 90%. The complete behavior-complexity gate also evaluates the unique executed-action count, both entropy diagnostics, and the predeclared condition-dependence criteria. These diagnostics include the seven artificial boundaries between independently reset windows. They support interpretation of the learned schedule; the primary performance claim remains the paired forecast loss comparison.

6. Results

The evaluation covers forecast accuracy and the executed schedules that produce it. Mean forecast loss and macro-averaged normalized forecast loss are co-primary endpoints. The validation-selected fixed-schedule baseline is the structural primary comparator, and the warning-rule baseline is the practical adaptive comparator. Scalar-objective and algorithm comparisons concern complete training configurations rather than isolated components.

6.1. *Forecasting performance on event-balanced evaluation windows*

Table 6 reports the primary paired analysis for the 22 post-selection evaluation seeds (119–140). The complete PD-PPO training configuration has lower mean forecast loss and lower macro-averaged normalized forecast loss than the validation-selected fixed-schedule baseline in all 22 seeds. Mean forecast loss decreases from 1.5744 to 1.4073, a relative reduction of 10.6%. The macro-averaged loss decreases from 1.0440 to 0.9602, a relative reduction of 8.0%. The mean paired differences against the fixed-schedule baseline are +0.1671 for mean forecast loss and +0.0838 for the macro-averaged loss, with 95% percentile bootstrap confidence intervals for the mean paired differences that remain above zero. The exact one-sided sign test probability for a direction fixed as lower loss, conditional on 22 independent signs under a 0.5 null, is 2.38×10^{-7} for either co-primary endpoint.

Seeds 117 and 118 supplied selected test-window metrics for a two-variant pilot architecture comparison, after which the selected configuration was frozen. They are excluded from the primary analysis. Supplementary Table S12 retains the 24-seed descriptive aggregate, including the 10.1% and 7.9% reductions, without assigning it post-selection inferential status. The PD-PPO scheduler also has lower values of both losses than the AoI-priority, round-robin, and random-feasible baselines in all 22 seeds.

The eight separately reset 512-step windows are constructed from test truth to balance particle, flux, and thermal coverage and emphasize transport intensity. The resulting event-balanced evaluation windows do not represent the natural test distribution. Supplementary Table S7 tests whether that sampling rule drives the result by evaluating the same final policy checkpoints and the fixed channel subsets selected on the calibration/validation partition once over all 5,242 scoreable test

Table 6. Baseline comparisons with the complete PD-PPO training configuration on the event-balanced evaluation windows over 22 post-selection evaluation seeds. A positive paired difference means that the baseline has higher loss than PD-PPO.

Baseline	Mean forecast loss Seeds with lower loss; mean paired difference [95% bootstrap CI]	Macro-averaged loss Seeds with lower loss; mean paired difference [95% bootstrap CI]
Fixed-schedule baseline	22 of 22; +0.1671 [+0.1231, +0.2164]	22 of 22; +0.0838 [+0.0708, +0.0967]
AoI-priority baseline	22 of 22; +0.2619 [+0.2331, +0.2928]	22 of 22; +0.1655 [+0.1430, +0.1898]
Round-robin baseline	22 of 22; +0.2571 [+0.2249, +0.2900]	22 of 22; +0.1664 [+0.1427, +0.1917]
Random-feasible baseline	22 of 22; +0.2681 [+0.2366, +0.3019]	22 of 22; +0.1684 [+0.1446, +0.1933]
Privileged one-step look-ahead reference using next-step test targets	22 of 22; +0.2746 [+0.2391, +0.3107]	22 of 22; +0.1855 [+0.1559, +0.2173]
Warning-rule baseline	8 of 22; -0.0039 [-0.0102, +0.0019]	9 of 22; +0.0011 [-0.0049, +0.0081]
Privileged true-label baseline	9 of 22; -0.0042 [-0.0108, +0.0011]	10 of 22; +0.0017 [-0.0045, +0.0086]

Each interval is a 95% bootstrap confidence interval for the paired difference. The validation-selected fixed-schedule baseline is the structural primary comparator; the warning-rule baseline is the practical adaptive comparator. The one-step look-ahead reference uses next-step test targets, and the true-label baseline uses the exact sequence $c_t, \dots, c_{t+16}$; both receive information unavailable during deployment, and neither is an upper bound. The warning-rule baseline uses the same simulator-generated warning signals available to PD-PPO and a mapping selected on the calibration/validation partition.

time steps in each continuous test partition. The PD-PPO scheduler again has lower mean forecast loss and lower macro-averaged loss in all 22 seeds. The mean paired differences are +0.1347 for mean forecast loss and +0.0836 for the macro-averaged loss, and both 95% percentile bootstrap confidence intervals for the mean paired differences remain above zero. The observed paired advantage therefore extends beyond the windows emphasizing transport periods used for the primary balanced comparison. Figure 3 summarizes the seed-level paired differences for the main comparison with the fixed-schedule baseline and the additional baselines.

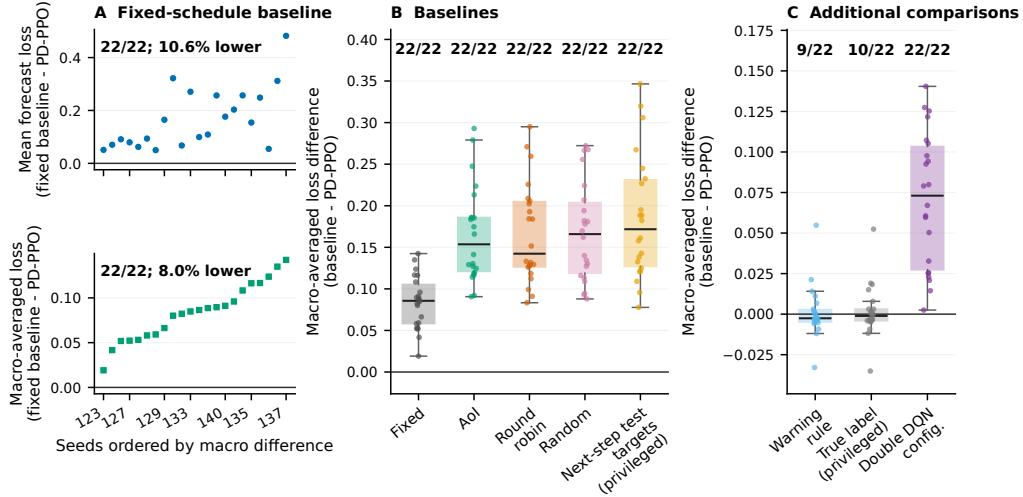


Figure 3. Seed-level paired differences over 22 post-selection evaluation seeds. Panel A compares PD-PPO with the fixed-schedule baseline. Panel B compares the AoI-priority, round-robin, and random-feasible baselines and the privileged one-step look-ahead reference using next-step test targets. Panel C compares the warning-rule and privileged true-label baselines and the Double DQN training configuration. Positive values mean that the baseline has higher loss than PD-PPO.

6.2. Channel activation across operating conditions

The aggregate gain is distributed across all three forecast event types (Table 7). Mean paired differences are positive for particle, flux, and thermal event types, and each 95% percentile bootstrap confidence interval for the mean paired difference remains above zero. The PD-PPO scheduler has lower loss in 18 of 22 particle comparisons, 12 of 22 flux comparisons, and 15 of 22 thermal comparisons. Equal weighting combines the three distinct specialist-channel requirements and yields lower macro-averaged loss in all 22 seeds.

Table 7. Event-type-specific forecast losses and the macro-averaged normalized forecast loss. Lower values are better; a positive paired difference favors PD-PPO over the fixed-schedule baseline.

Event type	PD-PPO	Fixed baseline	Seeds with lower loss	Mean paired difference [95% bootstrap CI]
Particle event type	1.0749	1.1037	18 of 22	+0.0287 [+0.0148, +0.0473]
Flux event type	0.7639	0.8930	12 of 22	+0.1291 [+0.0642, +0.1938]
Thermal event type	1.0418	1.1353	15 of 22	+0.0935 [+0.0444, +0.1452]
Macro-averaged loss	0.9602	1.0440	22 of 22	+0.0838 [+0.0708, +0.0967]

Figure 4 compares optional-channel activation by PD-PPO and the validation-selected fixed-schedule baseline. The laser disdrometer channel is active for 99.2% of particle-event steps, the FC4 snow mass-flux channel for 97.8% of flux-event steps, and the infrared snow-surface-temperature channel for 99.7% of thermal-event steps. Calm periods use a mixture of the four optional channels with nonzero test activation frequency. Calibration/validation selects the FC4 snow mass-flux channel in 11 seeds, the infrared snow-surface-temperature channel in nine, and the laser disdrometer channel in two, after which the selected subset remains fixed on the test partition.

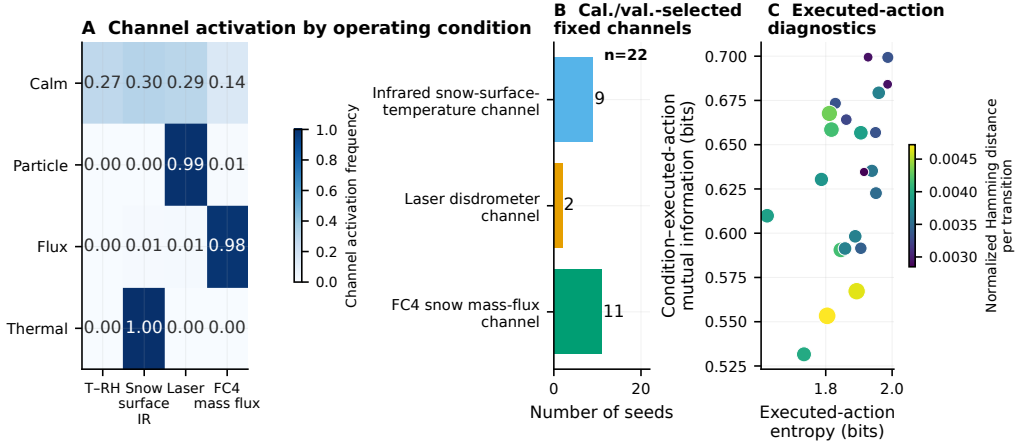


Figure 4. Optional-channel activation by operating condition. Panel A shows the shielded temperature–humidity, infrared snow-surface-temperature, laser disdrometer, and FC4 snow mass-flux channels. Panel B shows the fixed channels selected on the calibration/validation partition. Panel C reports executed-action entropy, condition–executed-action mutual information, and mean normalized Hamming distance between adjacent executed masks.

All 22 executed-action sequences pass the behavior-complexity gate; none is fixed-like or simple-cycle-like under its prespecified criteria. Mean executed-action entropy is 1.872 bits, and mean condition–executed-action mutual information is 0.632 bits. The mean normalized Hamming distance between adjacent executed masks is 0.00369. The meteorological core channel remains active by construction, while the executed schedules do not activate the basic radiometer channel in the selected test windows. At least three of the four active optional channels have intermediate channel activation frequency in every seed; all four do so in 21 of 22 seeds. No channel was deactivated before completing warm-up because $D_{\min} = 6$ exceeds the maximum two-step warm-up. Executed channel changes occur mainly when the operating condition changes. The gate’s fixed-like and simple-cycle-like criteria exclude fixed schedules and simple cycles. These behavior

diagnostics concatenate the eight separately reset windows and include the seven reset boundaries; they are descriptive diagnostics, not a primary performance endpoint.

6.3. *Scalar-objective and training-configuration comparisons*

The AoI-objective and uncertainty-objective PPO configurations retain the PPO architecture, scheduler input, feasible-action mask, supervised target actions used only during training, auxiliary future-condition classifier, chronological partitions, and frozen primary forecaster, but use AoI or a bounded diagonal uncertainty proxy as the scalar rollout objective. Because each objective supplies its own advantages, the comparison also changes the PPO surrogate, value target, and advantage-weighted behavior cloning weights. Table 8 reports the paired differences from PD-PPO. The mean macro-averaged loss difference is +0.0015 for the AoI configuration, with lower loss for the forecast-loss-objective configuration in 12 of 22 seeds, and +0.0013 for the uncertainty configuration, with lower loss in 12 of 22 seeds. For both alternative configurations, both 95% bootstrap confidence intervals for the paired difference in macro-averaged loss and mean forecast loss include zero. These comparisons establish neither a population-level directional difference nor equivalence among scalar objectives. All three matched PPO configurations nevertheless have lower macro-averaged loss than the fixed-schedule baseline in all 22 seeds and meet the channel-use acceptance criterion in every seed. Warm-up interruption is structurally zero for all three configurations because $D_{\min} = 6$ exceeds the maximum two-step warm-up.

The Double DQN training configuration uses the same scheduler observation, reward based on forecast loss, action set, constraints, policy-training partition, and frozen primary forecaster. PD-PPO has lower macro-averaged loss in all 22 seeds

and lower mean forecast loss in 21 of 22, with mean baseline-minus-PD-PPO paired differences of +0.0706 and +0.1423, respectively. Double DQN has lower macro-averaged loss than the fixed-schedule baseline in 12 of 22 comparisons and meets the channel-use acceptance criterion in 5 of 22 seeds; no channel can be deactivated before completing warm-up under the reported $D_{\min}$. All test metrics use the same frozen primary forecaster. This is a training-configuration comparison on the same scheduling problem, because the complete PD-PPO configuration uses behavior cloning pretraining, the continuing advantage-weighted behavior cloning term, and the auxiliary future-condition classifier; it is not an isolated comparison of the PPO and Double DQN update rules.

6.4. *Baselines using additional information*

The privileged one-step look-ahead reference using next-step test targets uses information unavailable during deployment. At each time step it reads next-step targets on the test partition and evaluates feasible actions with their next-step forecast loss. Its one-step scoring does not account for the longer-term consequences of the minimum dwell-time constraint, warm-up, and observation age. The PD-PPO scheduler has lower mean forecast loss and lower macro-averaged normalized forecast loss in all 22 seeds (Table 6). For this reference, the mean normalized Hamming distance between adjacent executed masks is 0.0280. Despite access to next-step forecast loss, it has higher mean forecast loss and higher macro-averaged normalized forecast loss in every seed, and its executed activation indicators change approximately 7.6 times as often as those of PD-PPO.

The warning-rule baseline maps simulated warning signals to proposed actions. The privileged true-label baseline uses the exact sequence $c_t, \dots, c_{t+16}$ and a condition-to-action mapping selected on the calibration/validation partition. These

Table 8. Scalar-objective PPO and Double DQN training-configuration comparisons. A positive paired difference indicates lower loss for PD-PPO.

Comparison	Mean forecast loss Seeds with lower loss; mean paired difference [95% bootstrap CI]	Macro-averaged loss Seeds with lower loss; mean paired difference [95% bootstrap CI]	Lower macro- averaged loss than fixed baseline	Channel-use acceptance criterion
AoI-objective PPO configuration	10 of 22; -0.0003 [-0.0073, +0.0063]	12 of 22; +0.0015 [-0.0037, +0.0066]	22 of 22	22 of 22
Uncertainty- objective PPO configuration	10 of 22; +0.0008 [-0.0063, +0.0076]	12 of 22; +0.0013 [-0.0035, +0.0054]	22 of 22	22 of 22
Double DQN training configuration	21 of 22; +0.1423 [+0.1036, +0.1827]	22 of 22; +0.0706 [+0.0535, +0.0875]	12 of 22	5 of 22

Each interval is a 95% bootstrap confidence interval. The objective rows compare matched PPO training configurations with different scalar objectives. Each objective changes the PPO surrogate, value target, advantage, and the weights in the advantage-weighted behavior cloning term; the rows are therefore not isolated scalar-reward ablations. The configurations otherwise share the scheduler input, training-only supervised targets, auxiliary future-condition classifier, candidate action set, and constraints. The Double DQN row is a training-configuration comparison: it shares the state representation, reward based on forecast loss, action set, and constraints, but omits behavior cloning pretraining, the continuing behavior cloning term, and the auxiliary classifier. All configurations have zero warm-up interruption structurally because $D_{\min} = 6$ exceeds the maximum two-step warm-up; the final column is therefore determined by the channel-use acceptance criterion. All test metrics use the same frozen primary forecaster.

condition labels are unavailable during deployment. Their mean macro-averaged loss differences from PD-PPO are +0.0011 and +0.0017, respectively. For both baselines, the 95% percentile bootstrap confidence intervals for the mean paired differences include zero for macro-averaged loss and mean forecast loss. These intervals establish neither a population-level directional difference nor equivalence. The true-label baseline is neither deployable nor an upper bound on feasible sequential control.

6.5. Alternative-forecaster analysis

The alternative-forecaster analysis evaluates the test observation sequences with a ridge forecaster. Fixed-schedule selection was repeated on the calibration/validation partition before test evaluation. PD-PPO has lower macro-averaged loss than the primary-forecaster fixed-schedule baseline in all 22 comparisons and than the ridge-forecaster fixed-schedule baseline in 21 of 22 (Supplementary Table S8). The mean paired difference against the fixed schedule selected using the ridge forecaster is +0.1339, with a 95% percentile bootstrap confidence interval for the mean paired difference of [+0.1103, +0.1563]. This evaluation changes both the forecaster architecture and the validation-selected fixed-schedule baseline without retraining the policy. Figure 5 summarizes the scalar-objective comparisons and the two ridge-forecaster comparisons.

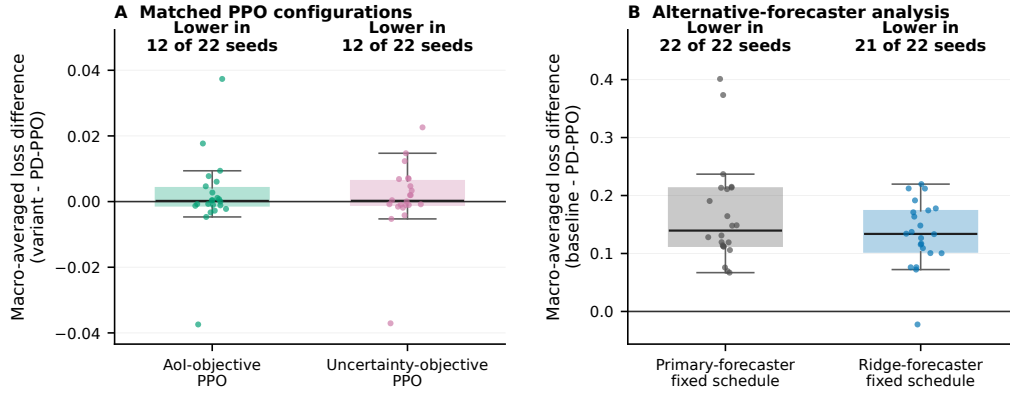


Figure 5. Scalar-objective training-configuration comparisons and alternative-forecaster analysis over 22 post-selection seeds. Panel A compares matched PPO configurations with different scalar objectives. Panel B gives macro-averaged loss differences for the primary-forecaster fixed-schedule baseline and the fixed schedule selected with the ridge forecaster. Positive values mean that the baseline has higher loss than PD-PPO.

7. Discussion

7.1. Primary comparison with the fixed-schedule baseline

Across the 22 post-selection evaluation seeds, the complete PD-PPO training configuration achieved lower mean forecast loss and lower macro-averaged normalized forecast loss than the validation-selected fixed-schedule baseline in every seed. The separate 24-seed descriptive aggregate, which includes two pilot/model-selection seeds, retains the same direction. With the ridge forecaster, the paired difference against the fixed baseline selected for that forecaster is positive in 21 of the 22 post-selection seeds and 23 of 24 seeds in the descriptive aggregate. Evaluation over all scoreable test steps also retains lower values of both co-primary endpoints in every seed, so the observed difference is not confined to the event-balanced evaluation windows. Executed-action analysis shows that the scheduler activates

the corresponding specialist channel for each of the particle, flux, and thermal event types while retaining a broader mixture during calm periods. This pattern was observed for the complete training configuration, which includes simulated warning inputs and future-condition supervision; it cannot be attributed to the scalar objective alone.

The reported system has a mandatory meteorological core channel and an optional-channel capacity of one. The underlying design problem extends to settings in which only $q < n_{\text{opt}}$ optional channels can operate. In such systems, a scheduler must decide whether to reserve the available capacity for one generally useful measurement or reallocate it as forecast requirements change. The present $q = 1$ optional-channel capacity makes that trade-off directly observable and prevents simultaneous low-cost optional channels. The meteorological core channel costs 0.25 normalized units and each event-specific channel costs 0.49. Although two event-specific channels would exceed the 0.75 budget, such subsets are absent from the candidate set. Every listed candidate has steady-state demand at most 0.74 and startup-peak demand at most 0.92, below the corresponding 0.75 and 0.95 limits. Both instantaneous power limits are enforced, but the feasible-action mask therefore excludes no candidate in this instance. The empirical result is established under the optional-channel cardinality constraint and minimum dwell-time constraint, not under observed binding of the two power limits (Supplementary Table S14).

7.2. *Condition-dependent activation relative to the fixed schedule*

Calibration/validation selects a strong fixed optional channel. The FC4 snow mass-flux channel is selected in more than half of the seeds, and the infrared snow-surface-temperature channel in most of the remainder. Such a schedule avoids switching and can perform well when the evaluation distribution is dominated

by one measurement requirement. Its weakness appears when particle, flux, and thermal event types all contribute to the forecasting objective. The optional channel preferred for one event type is nearly absent from another, so no single subset can preserve the three event-specific benefits. This pattern instantiates the proposition’s motivation, but the empirical table does not verify all of its policy-by-condition assumptions.

The PD-PPO scheduler obtains the observed difference with sparse executed-action changes. The mean normalized Hamming distance per transition is 0.00369, consistent with the minimum dwell-time constraint and below the corresponding rate for the one-step look-ahead reference under its one-step rule. Executed changes mainly coincide with changes in operating condition, and no optional channel dominates the sequence across event types. Nonzero condition–executed-action mutual information and the behavior-complexity gate distinguish these executed sequences from fixed-like or simple-cycle-like schedules under the frozen thresholds. These quantities are descriptive and include boundaries between the eight separately reset event-balanced evaluation windows.

7.3. Scalar-objective and training-configuration comparisons

The matched AoI-objective and uncertainty-objective PPO configurations replace forecast loss with AoI or a bounded diagonal uncertainty proxy while retaining the shared PPO algorithm. Proposition 2 establishes a mathematical distinction among the theoretical forecast-loss, AoI, and covariance-trace objectives; the empirical uncertainty configuration uses a bounded diagonal proxy rather than the full covariance trace. All three PPO configurations receive the same condition-specific training weights, training-only supervised target actions, and auxiliary future-condition classification labels. Each scalar objective also changes the value

target, advantage, PPO surrogate, and advantage weights used in the continuing behavior cloning term. The feasible action set is small, and four optional-channel choices have nonzero activation frequency under the main policy. The 95% percentile bootstrap confidence intervals for the mean paired differences include zero for both matched scalar-objective PPO configurations. These comparisons concern matched PPO training configurations with different scalar objectives; they are not reward-only ablations.

The Double DQN training configuration uses the same scheduler observation, reward based on forecast loss, action set, and constraints. It has higher macro-averaged loss in all 22 post-selection seeds and higher mean forecast loss in 21 of 22 seeds than PD-PPO. Under the channel-use acceptance criterion, Double DQN meets the criterion in 5 of 22 seeds; no channel is deactivated before warm-up completes, as structurally required by $D_{\min} = 6$. The PD-PPO configuration also contains behavior cloning pretraining, continuing advantage-weighted behavior cloning, and an auxiliary future-condition classifier. The comparison therefore concerns complete training configurations and does not establish a general advantage of policy-gradient learning over value-based learning.

The privileged one-step look-ahead reference using next-step test targets can inspect next-step forecast loss on the test partition and therefore has more information per decision than the deployed scheduler. It still has higher loss in every seed and a mean normalized Hamming distance about 7.6 times as large. The minimum dwell-time constraint, warm-up delay, and persistence of stale observations couple consecutive actions, but the reference objective omits their longer-term consequences. The result shows that next-step target access under this one-step rule does not yield lower multi-step loss; it does not isolate the value of sequential decision

making or compare PD-PPO with a constrained model-predictive controller.

7.4. *Effects of warning signals and true labels*

The warning-rule baseline and privileged true-label baseline test scheduler behavior under different sources of condition-related information. The simulated warning signals are leading scores generated from the simulated operating-condition process. The warning-rule baseline applies the predeclared threshold and a condition-to-action mapping selected on the calibration/validation partition. The privileged true-label baseline removes warning uncertainty by using the exact sequence $c_t, \dots, c_{t+16}$. Neither paired confidence interval excludes zero; the intervals establish neither a population-level directional difference nor equivalence. The PD-PPO scheduler instead combines simulated warning signals with partial observations, channel sampling ages, channel operating states, and the previous executed channel subset.

Both baselines depend on explicit condition-to-action mappings selected on the calibration/validation partition. The true-label baseline uses condition labels that are unavailable during deployment. The policy selects a proposed action index from the feasible index set using current observations, channel operating states, and recent-history features, and can retain information from observations between warning transitions. Robustness to warning quality remains untested. Over the continuous scoreable test partitions of the 22 post-selection seeds, the thresholded four-class warning rule has 0.909 accuracy, a 0.916 macro-averaged F1 score, and 1.471 bits of mutual information with the current condition label. Subtype-wise one-vs-rest ROC AUC is 0.997–0.999, and all 1,262 event onsets cross the 0.5 threshold within the preceding 16 steps, with median lead time nine steps. These diagnostics establish that the simulator-generated warning features

are highly informative within the reported simulation; they do not validate a field detector or test weak-warning conditions. Supplementary Table S13 gives the component metrics and analysis scope.

7.5. Online computation and scaling

The frozen primary forecaster is used for policy training and offline evaluation. Online execution requires one policy forward pass followed by a feasibility check and one check of the minimum dwell-time constraint. It neither simulates future observation sequences nor evaluates every candidate with a forecaster. In the six-channel system, the categorical policy scores six candidate indices; no runtime benchmark against instrument sampling and estimation is reported.

With several simultaneously active optional channels, explicit enumeration can grow combinatorially. A larger action space could retain the downstream forecast-loss objective while requiring a different feasible-action parameterization, such as sequential channel selection or direct feasible-subset construction. The appropriate parameterization depends on the number of channels and on whether several channel combinations carry distinct forecast value.

7.6. Limitations

The study uses a controlled simulation for cold-region monitoring, including simulated warning signals and six logical channels. The experiment was designed with three event types, three corresponding specialist channels, prescribed target weights, and leading warnings. It tests whether a scheduler recovers a designed condition-dependent sensing allocation under the specified information and execution structure. The 22 post-selection seeds and two pilot seeds cover new event realizations from one generator, not different sites, hardware assemblies, power

systems, channel configurations, or warning systems. Field use requires measured power and warm-up profiles, warning signals derived from available instruments, and evaluation under real missingness and faults. The privileged true-label baseline uses evaluation-only information and is not part of the scheduler input.

The macro-averaged normalized forecast loss uses event-balanced evaluation windows constructed from test truth using event coverage and transport intensity so that all three event-specific measurement requirements receive explicit support. Evaluation over all scoreable time steps in the test partition retains the paired difference between the complete PD-PPO training configuration and the validation-selected fixed-schedule baseline, but it does not replace evaluation across different sites or event-generation processes outside the simulation model. The reported result is established for mean forecast loss and macro-averaged normalized forecast loss; invariance to alternative target or event-type weightings is not claimed.

The primary comparison uses the frozen primary forecaster. The alternative-forecaster analysis uses a separately fitted, frozen ridge forecaster to evaluate existing observation sequences without policy retraining. The scalar-objective comparisons also retain behavior cloning pretraining, continuing advantage-weighted behavior cloning, and auxiliary future-condition classification labels. The reported implementation stores sampled proposed-action log probabilities even when the previous action is executed, and the proposal-override rate was not logged; the result therefore supports the complete implementation rather than an idealized executed-action likelihood convention. Both power limits are non-excluding, and the warm-up-interruption penalty is structurally zero in the reported instance. Further work should test hardware, other channel sets, warning quality, and joint scheduler–forecaster adaptation.

8. Conclusion

The PD-PPO scheduler formulates sensor scheduling under power constraints as downstream forecast optimization over feasible channel subsets. The policy uses the scheduler observation to select a proposed action index, the feasible-action mask enforces the steady-state and startup-peak power limits, and the scheduler applies the minimum dwell-time constraint before executing an activation vector. In the six-channel simulation, across the 22 post-selection evaluation seeds, the complete PD-PPO training configuration achieved 10.6% lower mean forecast loss and 8.0% lower macro-averaged normalized forecast loss than the validation-selected fixed-schedule baseline. Both co-primary endpoints are lower in every post-selection seed. The 24-seed descriptive aggregate, which includes the two pilot/model-selection seeds, retains the previously reported 10.1% and 7.9% reductions. Evaluation over all 5,242 scoreable test time steps preserves this direction for both co-primary endpoints. The executed schedules use optional channels condition-dependently; all 22 executed action sequences pass the behavior-complexity gate.

The complete PD-PPO training configuration has lower macro-averaged loss than the Double DQN training configuration in all 22 post-selection seeds and lower mean forecast loss in 21. The configurations also differ in behavior cloning pretraining, continuing advantage-weighted behavior cloning, and auxiliary future-condition classification, so this is a comparison of complete training configurations. The paired difference against the fixed-schedule baseline remains positive under the ridge forecaster. For each of the AoI-objective and uncertainty-objective PPO configurations, the 95% bootstrap confidence intervals for both co-primary paired differences include zero. The corresponding intervals against the warning-rule and privileged true-label baselines also include zero; these intervals establish

neither a population-level directional difference nor equivalence. The privileged true-label baseline uses the exact sequence $c_t, \dots, c_{t+16}$, which is unavailable during deployment. In this controlled simulation, the complete PD-PPO training configuration produces condition-dependent executed channel activation and lower forecast loss than the validation-selected fixed-schedule baseline. The evidence does not isolate the scalar objective, PPO update, highly informative warning signals, or training-only supervision.

Appendix A. Supplementary Theoretical Details

Appendix A.1. Proof of Proposition 1

Let $\tilde{L}_c(\pi)$ denote the normalized condition-specific loss introduced before the proposition. For $s \in \mathcal{S}$, a fixed policy π_s chooses the same optional channel s for every operating condition. The policy π_{label} has privileged access to the condition label and changes to the corresponding optional channel whenever the common startup and minimum dwell-time constraints permit. Condition intervals are long enough for these changes, and the losses incurred during transitions are included in $\tilde{L}_c$.

For a fixed optional channel s , define $\delta_c(s) = \tilde{L}_c(\pi_s) - \tilde{L}_c(\pi_{\text{label}})$ and the mismatched-condition set

$$\mathcal{C}_{\text{mis}}(s) = \{c \in \mathcal{C} \mid s \text{ mismatches the best optional channel in } c\}. \quad (\text{A.1})$$

By assumption, $\delta_c(s) \geq 0$ for every condition and $\delta_c(s) \geq \Delta_c > 0$ for each mismatch. The weighted normalized forecast loss of the fixed channel therefore satisfies

$$L_\rho(\pi_s) - L_\rho(\pi_{\text{label}}) = \sum_{c \in \mathcal{C}} \rho_c \delta_c(s) \geq \sum_{c \in \mathcal{C}_{\text{mis}}(s)} \rho_c \Delta_c > 0. \quad (\text{A.2})$$

The strict inequality follows because $C_{\text{mis}}(s)$ is non-empty and every included term is positive. Thus every fixed channel has higher weighted normalized forecast loss under the stated sufficient conditions.

Appendix A.2. Proof of Proposition 2

Set the common objective horizon to $T = 1$; then all three objectives assign the same temporal weight $\gamma^0 = 1$ to the evaluated step. Consider one forecast target, one forecast horizon, and two independent scalar Gaussian random walks

$$z_{t+1} = z_t + \xi_t, \quad e_{t+1} = e_t + \zeta_t, \quad (\text{A.3})$$

where $\xi_t \sim \mathcal{N}(0, q_z)$ and $\zeta_t \sim \mathcal{N}(0, q_e)$ are mutually independent over time. The one-step forecast target is

$$y_{t+1} = cz_t + Me_t + v_{t+1}, \quad (\text{A.4})$$

with independent $v_{t+1} \sim \mathcal{N}(0, \sigma_v^2)$ and coefficients $M > c > 0$. Suppose that one random-walk component can be measured without noise at each time step. At the evaluated time step, feasible schedules can produce age vectors $d^{(1)} = (0, 1)$ under policy π_1 and $d^{(2)} = (2, 0)$ under policy π_2 . Here $\mathcal{Y} = \{y\}$ for the forecast-loss objective and $\mathcal{V}_{\text{AoI}} = \{z, e\}$ for the AoI objective. These vectors can arise from feasible one-measurement schedules that prioritize different random-walk components before the evaluated time step. Thus π_1 has lower total AoI because $1 < 2$.

The corresponding posterior error covariances are $P^{(1)} = \text{diag}(0, q_e)$ and $P^{(2)} = \text{diag}(2q_z, 0)$. Choose $q_z > 0$ and

$$\frac{2c^2q_z}{M^2} < q_e < 2q_z, \quad (\text{A.5})$$

which is possible because $M > c$. The right inequality gives $\text{tr } P^{(1)} < \text{tr } P^{(2)}$, so the covariance trace objective also prefers π_1 .

Now take the single-target, single-horizon, positive-weight special case of Equation (5). Its positive weight cancels between the numerator and denominator, so the per-step forecast loss is

$$L_{\text{step}}(t) = \min \left\{ 10, \frac{|\widehat{y}_{t+1} - y_{t+1}|}{\sigma} \right\}, \quad \sigma > 0. \quad (\text{A.6})$$

The conditional forecast errors under π_1 and π_2 are zero-mean Gaussian with variances

$$v_1 = M^2 q_e + \sigma_v^2, \quad v_2 = 2c^2 q_z + \sigma_v^2. \quad (\text{A.7})$$

The left inequality in Equation (A.5) gives $v_1 > v_2$.

For $v > 0$, let $Z_v \sim \mathcal{N}(0, v)$ and define

$$g(v) = \mathbb{E} \left[\min \left\{ 10, \frac{|Z_v|}{\sigma} \right\} \right]. \quad (\text{A.8})$$

Writing $Z_v = \sqrt{v} U$ with $U \sim \mathcal{N}(0, 1)$ shows that the integrand for v_1 is at least that for v_2 at every U . The inequality is strict on

$$0 < |U| < \frac{10\sigma}{\sqrt{v_1}}, \quad (\text{A.9})$$

which has positive probability. Hence $g(v_1) > g(v_2)$, and

$$\mathbb{E}[L_{\text{step}}(t; \pi_1)] > \mathbb{E}[L_{\text{step}}(t; \pi_2)]. \quad (\text{A.10})$$

Because $T = 1$ in Equation (8), this gives $J_{\text{fcst}}(\pi_2) > J_{\text{fcst}}(\pi_1)$. The AoI and covariance trace objectives therefore prefer π_1 , while the capped, scale-normalized forecast-loss objective prefers π_2 .

Appendix A.3. Observation-dependent scheduling and fixed cycles

A fixed cyclic schedule chooses optional channels from the time index alone. Such a schedule may be optimal in structured settings, including periodic conditions or rewards that are phase-aligned with the cycle. Without that structure, a fixed cycle need not be optimal. If informative observations predict a change in operating condition and a cycle has nonzero probability of assigning a mismatched optional channel, the positive mismatch loss condition in Proposition 1 yields positive expected regret relative to a policy conditioned on those observations. Action sequences are therefore needed to distinguish observation-dependent scheduling from a fixed schedule or a fixed cyclic schedule.

Appendix A.4. Relation to fixed channel-subset allocation

Proposition 2 does not imply that an adaptive policy must beat a fixed channel subset. If one fixed optional channel measures the variables that dominate a forecast metric, a fixed policy can remain optimal for that metric. Adaptive value requires the incompatible-condition requirement in Proposition 1, together with observations that provide useful information about a change in operating condition. Mean forecast loss and weighted normalized forecast loss can therefore rank a fixed and an adaptive policy differently.

Appendix B. Simulation Validation

Table B.9 reports the acceptance checks used to bound the simulation domain. Supplementary Fig. S3 provides the corresponding distribution, autocorrelation, and event-window plots. These checks document the simulation construction and are not policy-selection criteria.

Table B.9. Simulation validation checks and acceptance criteria.

Validation criterion	Value	Acceptance rule	Passed
Wind-speed autocorrelation max deviation, lags 1–12 h	0.048	< 0.05	yes
$P(\text{event fraction} > 0.75)$ in 512-step windows	0.065 [0.062, 0.067]	> 0.05	yes
$P(\text{event fraction} < 0.25)$ in 512-step windows	0.539	> 0.30	yes
Max KS statistic against reference distributions from Antarctic AWS stations	0.0146	< 0.05	yes
Median blowing-snow event duration	18.0 h	12–20 h	yes
Flux–wind relationship, log–log slope	2.966	2.5–3.5	yes
Particle-size–wind correlation, Spearman	–0.544	< -0.30	yes
Wind-speed power spectrum log-MSE, 0.1–4.0 cpd	0.0419	< 0.10	yes

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

A versioned repository containing the code, aggregate tables, seed summaries, generated figure assets, and reproduction scripts will be made publicly available upon acceptance. An anonymous review archive contains the code and evidence needed to verify the reported aggregate results. All reported empirical evidence comes from simulated data; field measurements are outside the evaluation set.

Declaration of Generative AI and AI-Assisted Technologies

During preparation of this manuscript the authors used OpenAI Codex and Hermes Agent for code inspection, consistency checks, and language-editing assistance. The authors reviewed and edited all generated material and take full responsibility for the content of the manuscript. No figure or other graphical asset in the manuscript was created or modified using a generative image tool.

References

- Ahdab, M.A., Leth, J.J., Tan, Z.H., 2025. Optimal sensor scheduling and selection for continuous-discrete Kalman filtering with auxiliary dynamics, in: Proc. 42nd Int. Conf. Mach. Learn. (ICML 2025), PMLR. pp. 703–729. URL: <https://proceedings.mlr.press/v267/ahdab25a.html>.
- Alali, M., Kazeminajafabadi, A., Imani, M., 2024. Deep reinforcement learning sensor scheduling for effective monitoring of dynamical systems. Syst. Sci. Control Eng. 12, 2329260. doi:[10.1080/21642583.2024.2329260](https://doi.org/10.1080/21642583.2024.2329260).
- Aloni, O., Perelman, G., Fishbain, B., 2025. Synthetic random environmental time series generation with similarity control, preserving original signal’s statistical characteristics. Environmental Modelling & Software 185, 106283. doi:[10.1016/j.envsoft.2024.106283](https://doi.org/10.1016/j.envsoft.2024.106283).
- Amory, C., 2020. Drifting-snow statistics from multiple-year autonomous measurements in Adélie Land, East Antarctica. The Cryosphere 14, 1713–1725. doi:[10.5194/tc-14-1713-2020](https://doi.org/10.5194/tc-14-1713-2020).
- Bai, S., Kolter, J.Z., Koltun, V., 2018. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv preprint arXiv:1803.01271 doi:[10.48550/arXiv.1803.01271](https://doi.org/10.48550/arXiv.1803.01271), [arXiv:1803.01271](https://arxiv.org/abs/1803.01271).
- Bajcsy, R., Aloimonos, Y., Tsotsos, J.K., 2018. Revisiting active perception. Auton. Robots 42, 177–196. doi:[10.1007/s10514-017-9615-3](https://doi.org/10.1007/s10514-017-9615-3).
- Fernández-Bes, J., Cid-Sueiro, J., Marques, A.G., 2015. An MDP model for censoring in harvesting sensors: Optimal and approximated solutions. IEEE J. Sel. Areas Commun. 33, 1717–1729. doi:[10.1109/JSAC.2015.2391792](https://doi.org/10.1109/JSAC.2015.2391792).

- Golovin, D., Krause, A., 2011. Adaptive submodularity: Theory and applications in active learning and stochastic optimization. *J. Artif. Intell. Res.* 42, 427–486. doi:[10.1613/jair.3278](https://doi.org/10.1613/jair.3278).
- Jonah, S., Yoo, S.K., Sthapit, S., 2026. Adaptive scheduling: A reinforcement learning Whittle index approach for wireless sensor networks. *IEEE Access* 14, 40042–40059. doi:[10.1109/ACCESS.2026.3673220](https://doi.org/10.1109/ACCESS.2026.3673220).
- Kaul, S., Yates, R., Gruteser, M., 2012. Real-time status: How often should one update?, in: 2012 Proceedings IEEE INFOCOM, IEEE. pp. 2731–2735. doi:[10.1109/INFCOM.2012.6195689](https://doi.org/10.1109/INFCOM.2012.6195689).
- Lauri, M., Hsu, D., Pajarinen, J., 2023. Partially observable Markov decision processes in robotics: A survey. *IEEE Trans. Robot.* 39, 21–40. doi:[10.1109/TRO.2022.3200138](https://doi.org/10.1109/TRO.2022.3200138).
- Lim, B., Arık, S.O., Loeff, N., Pfister, T., 2021. Temporal fusion transformers for interpretable multi-horizon time series forecasting. *Int. J. Forecast.* 37, 1748–1764. doi:[10.1016/j.ijforecast.2021.03.012](https://doi.org/10.1016/j.ijforecast.2021.03.012).
- Liu, Y., Hu, T., Zhang, H., Wu, H., Wang, S., Ma, L., Long, M., 2024. iTransformer: Inverted transformers are effective for time series forecasting, in: International Conference on Learning Representations, pp. 11116–11140. URL: https://proceedings.iclr.cc/paper_files/paper/2024/hash/2ea18fdc667e0ef2ad82b2b4d65147ad-Abstract-Conference.html.
- Monrad-Krohn, L.J., Buhren, C., Eickelmann, L., Foth, A., Frey, M.M., Maahn, M., Marie-Sainte, W., Mech, M., Poinot, T., Walbröl, A., 2026. Measurements of blowing snow particle size distribution using an open-path optical snow

- particle counter in Ny-Ålesund during winter 2025 [dataset]. PANGAEA. doi:[10.1594/PANGAEA.992701](https://doi.org/10.1594/PANGAEA.992701).
- Murad, A., Kraemer, F.A., Bach, K., Taylor, G., 2020. Information-driven adaptive sensing based on deep reinforcement learning, in: Proceedings of the 10th International Conference on the Internet of Things, ACM. pp. 1–8. doi:[10.1145/3410992.3411001](https://doi.org/10.1145/3410992.3411001).
- Ogbodo, C.O., Rogers, T.J., Dal Borgo, M., Wagg, D.J., 2026. Adaptive sensor steering strategy using deep reinforcement learning for dynamic data acquisition in digital twins. Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences 482, 20250326. doi:[10.1098/rspa.2025.0326](https://doi.org/10.1098/rspa.2025.0326).
- Pendyala, A., Atamna, A., Glasmachers, T., 2024. Solving a real-world optimization problem using proximal policy optimization with curriculum learning and reward engineering, in: Bifet, A., Krilavičius, T., Miliou, I., Nowaczyk, S. (Eds.), Machine Learning and Knowledge Discovery in Databases. Applied Data Science Track: European Conference, ECML PKDD 2024, Vilnius, Lithuania, September 9–13, 2024, Proceedings, Part X. Springer Nature Switzerland, Cham. volume 14950 of *Lecture Notes in Computer Science*, pp. 150–165. doi:[10.1007/978-3-031-70381-2_10](https://doi.org/10.1007/978-3-031-70381-2_10).
- Qu, Z., Zhao, X., Xu, H., Tang, H., Wang, J., Li, B., 2022. An improved Q-learning-based sensor-scheduling algorithm for multi-target tracking. Sensors 22, 6972. doi:[10.3390/s22186972](https://doi.org/10.3390/s22186972).
- Sharma, V., Gerber, F., Lehning, M., 2023. Introducing CRYOWRF v1.0: multiscale

- atmospheric flow simulations with advanced snow cover modelling. *Geoscientific Model Development* 16, 719–749. doi:[10.5194/gmd-16-719-2023](https://doi.org/10.5194/gmd-16-719-2023).
- Shi, L., Cheng, P., Chen, J., 2011. Sensor data scheduling for optimal state estimation with communication energy constraint. *Automatica* 47, 1693–1698. doi:[10.1016/j.automatica.2011.02.037](https://doi.org/10.1016/j.automatica.2011.02.037).
- Tran, N.D., Mahmood, A., Gidlund, M., 2026. Learning-based sensor scheduling for delay-aware and stable remote state estimation. *arXiv preprint arXiv:2601.21482* doi:[10.48550/arXiv.2601.21482](https://doi.org/10.48550/arXiv.2601.21482).
- Wang, Y., Zhang, X., Ning, W., Lazzara, M.A., Ding, M., Reijmer, C.H., Smeets, P.C.J.P., Grigioni, P., Heil, P., Thomas, E.R., Mikolajczyk, D., Welhouse, L.J., Keller, L.M., Zhai, Z., Sun, Y., Hou, S., 2023. The AntAWS dataset: a compilation of Antarctic automatic weather station observations. *Earth Syst. Sci. Data* 15, 411–429. doi:[10.5194/essd-15-411-2023](https://doi.org/10.5194/essd-15-411-2023).
- Wei, Y., Zheng, R., 2020. Informative path planning for mobile sensing with reinforcement learning, in: *IEEE INFOCOM 2020 – IEEE Conference on Computer Communications*, IEEE. pp. 864–873. doi:[10.1109/INFOCOM41043.2020.9155528](https://doi.org/10.1109/INFOCOM41043.2020.9155528).